Supplementary Materials for

Evolutionary history from genomes to single cells by lossless compression

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Materials and Methods

Two-Part Minimum Description Length Objective and Star-Tree Null Baseline

In probabilistic phylogenetics, a substitution model and tree T assign a joint likelihood $P(A \mid T)$ to a discrete character matrix or multiple sequence alignment $A \in \Sigma^{N\times L}$ over N taxa (or single cells) and L columns on alphabet Σ (*2*). By Shannon's source coding theorem, $-\ln P(A \mid T)$ equals the exact code length in nats ($1 \text{ nat} = 1/\ln 2 \approx 1.4427$ bits) required to transmit A conditional on T. Standard model-selection criteria such as AIC and BIC (*10*, *63–65*) penalize the 1 to 10 continuous parameters of a rate matrix Q while leaving the $(N-3)$ discrete bipartition choices of T unpriced (*66*, *67*). Under Rissanen's two-part Minimum Description Length (MDL) principle (*7*, *8*), both the evolutionary graph $\mathcal{G} = (V, E_{\text{tree}}, E_{\text{shunt}})$ and the data encoded along it share one unit: $\mathcal{L}_{\text{MDL}}(A, \mathcal{G}) = \mathcal{L}(\mathcal{G}) + \mathcal{L}(A \mid \mathcal{G})$.

To establish an assumption-free null baseline, we price A on an unresolved star phylogeny T_{star} containing zero internal splits. Sending raw ungapped entries independently costs $N L_{\text{eff}} \ln|\Sigma|$ nats, whereas sending one ancestral root vector $X_r \in \Sigma^L$ costs $L \ln|\Sigma|$ nats plus, on each terminal branch $i \in \{1, \ldots, N\}$, only the m_i positions where lineage i differs across its $L_{\text{eff},i}$ observed columns:

$$\mathcal{L}_{\text{branch}}(m_i, L_{\text{eff},i}) \;=\; \mathcal{L}_{\mathbb{N}}(m_i) + \ln \binom{L_{\text{eff},i}}{m_i} + \sum_{k=1}^{m_i} \ell_{\text{subst}}(a_{i,k} \to b_{i,k}), \tag{1}$$

where $\mathcal{L}_{\mathbb{N}}(0) = \ln 2$ and, for $m_i \geq 1$,

$$\mathcal{L}_{\mathbb{N}}(m_i) \;=\; -\ln\left[\frac{\ln 2}{2}\left(\frac{1}{\ln(m_i+1)} - \frac{1}{\ln(m_i+2)}\right)\right]$$

is the Kraft-tight Elias–Rissanen universal integer code (*8*) satisfying $\sum_{m=0}^{\infty} e^{-\mathcal{L}_{\mathbb{N}}(m)} = 1$ identically without any separate active-branch indicator bit; $\ln\binom{L_{\text{eff},i}}{m_i}$ indexes which unordered subset of m_i columns mutated (vanishing at $m_i = 0$); and $\ell_{\text{subst}}(a \to b) = -\ln P(b \mid a)$ is the Kraft-normalized substitution surprise ($\sum_{b \neq a} e^{-\ell_{\text{subst}}(a \to b)} = 1$). Summing Eq. (1) over X_r and all leaves yields $\mathcal{L}_{\text{MDL}}(A, T_{\text{star}})$ and the dimensionless Phylogenetic Compression Ratio:

$$\rho(A) \;=\; 1 - \frac{\mathcal{L}_{\text{MDL}}(A, \mathcal{G}^*)}{\mathcal{L}_{\text{MDL}}(A, T_{\text{star}})}. \tag{2}$$

Exact Bipartition Cost, Rate Weighting, and the Natural Polytomy Law

Splitting a multifurcating hub w of degree $d = k_u + k_v \geq 4$ by inserting branch $e = (u, v)$ partitions its d incident lineages into $2 \leq k_u \leq k_v$ children. Indexing the unordered split size $\{k_u, k_v\}$ among $\max(1, \lfloor d/2 \rfloor - 1)$ non-trivial choices and selecting the k_u-subset (halved by symmetry when $k_u = k_v$) incurs the exact topological address cost:

$$\mathcal{L}_{\text{topo}}(k_u, k_v) \;=\; \ln \max\left(1,\; \left\lfloor \frac{k_u + k_v}{2} \right\rfloor - 1\right) \;+\; \ln \binom{k_u + k_v}{k_u} \;-\; \mathbb{I}(k_u = k_v) \ln 2 \quad \text{nats.} \tag{3}$$

For a quartet ($k_u = k_v = 2$), $\mathcal{L}_{\text{topo}}(2, 2) = \ln 3 \approx 1.099$ nats; within a degree-d burst hub, $\mathcal{L}_{\text{topo}}(2, d-2) \approx 3 \ln d - 2 \ln 2$ automatically corrects for searching across all $\binom{d}{2}$ pairs (unlike purely local edge tests (*68*, *69*)). Shifting one shared derived edit off an adjacent branch c carrying m_c mutations reduces its positional subset cost by

$$w_c \;=\; \ln \binom{L_{\text{eff}}}{m_c + 1} - \ln \binom{L_{\text{eff}}}{m_c} \;=\; \ln\left(\frac{L_{\text{eff}} - m_c}{m_c + 1}\right) \;\approx\; -\ln\left(\frac{p_c}{1 - p_c}\right) \quad \text{nats,} \tag{4}$$

recovering Felsenstein's exact maximum-likelihood log-odds branch weight (*2*). Writing $\Delta\mathcal{L}_{\text{children}}(m)$ for the child-branch code savings when m shared edits are factored onto e, split e is retained if and only if its net MDL compression gain is positive:

$$G_{\text{MDL}}(e) \;\equiv\; -\Delta\mathcal{L}_{\text{MDL}}(e) \;=\; \max_{1 \leq m \leq K} \left[\Delta\mathcal{L}_{\text{children}}(m) - \mathcal{L}_{\text{branch}}(m, L_{\text{eff}})\right] - \mathcal{L}_{\text{topo}}(k_u, k_v) \;>\; 0, \tag{5}$$

which maps directly onto a conditional local posterior branch probability $P_{\text{local}}(e \mid T_{-e}, A) = (1 + \exp(-G_{\text{MDL}}(e)))^{-1}$ comparing the locally optimal split state against collapsing e on the fixed surrounding topology T_{-e} without bootstrap resampling (*9*).

Four-Stage Hybrid Compression and Graph Induction Architecture

Finding the global MDL minimum across all $(2N - 5)!!$ leaf-labeled topologies is NP-hard, so Bushwhacker executes a deterministic four-stage search combining sub-quadratic distance/Voronoi seeding, vectorized continuous-time (`GTR+CAT8/LG08+CAT8`) or discrete/ordered (`-discrete`/`-ordered`) local optimization, post-hoc MDL contraction, and lateral shunt peeling:

1. **Bit-Parallel Encoding and Empirical Exchange Costs:** Nucleotide states and IUPAC ambiguities pack into four 64-bit integer bitplanes (`[u64; 4]`) per 64 alignment columns; variable-site patterns deduplicate deterministically by run-length encoding. Empirical transition ($\hat{p}_{\text{ti}}$) and transversion ($\hat{p}_{\text{tv}} = 1 - \hat{p}_{\text{ti}}$) costs ($\ell_{\text{ti}} = -\ln \hat{p}_{\text{ti}}$, $\ell_{\text{tv}} = -\ln(\hat{p}_{\text{tv}}/2)$) and empirical GTR/LG08 Markov parameters parameterize the discrete ledger and continuous-time likelihood sweeps.

2. **Stage 1 — Sub-Quadratic Starting Scaffold (T_0):** On a stratified pilot subsample ($K_{\text{sub}} \le 128$), sitewise Fitch edit counts s_k (*1*) under Poisson saturation $m_k = -k_{\text{sat}} \ln(1 - \min(0.92, s_k/k_{\text{sat}}))$ estimate the Gamma rate-dispersion shape $\hat{\alpha} \in [0.15, 50.0]$ and Jin–Nei ($\text{K80} + \Gamma(\hat{\alpha})$) distances (*70*):

$$\begin{aligned} \hat{\alpha} &= \frac{\bar{m}^2}{\max(10^{-6},\ \text{Var}(m) - \bar{m})}, \\ M_{ij} &= L_{ij} \cdot \frac{\hat{\alpha}}{2}\left[(1 - 2P_{ij} - Q_{ij})^{-1/\hat{\alpha}} + \frac{1}{2}(1 - 2Q_{ij})^{-1/\hat{\alpha}} - \frac{3}{2}\right]. \end{aligned} \tag{6}$$

   For $N \le 4{,}096$, BioNJ builds the binary starting scaffold T_0 from the distance matrix; for $N > 4{,}096$, Voronoi landmark divide-and-conquer partitions taxa and builds an $\mathcal{O}(N \log N \cdot L/64)$ skeleton (*71*).

3. **Stage 2 — Continuous-Time Rate-Category and Discrete Topology Refinement:** On all default nucleotide ($N \ge 4$) and protein/translated ($N \ge 3$) alignments, Bushwhacker assigns each variable column s unconditionally by nearest log-rate to one of eight unit-mean $\Gamma(1,1)$ octile rate categories ($r_1, \ldots, r_8,\ \frac{1}{8}\sum_{k=1}^{8} r_k = 1$) and runs a fixed deterministic schedule (`NNI + LeafMLE + SPR2 + NNI`) over 64-pattern SoA cache tiles (`GTR+CAT8` or `LG08+CAT8`): local `NNI` swaps accept Fisher-curvature-penalized log-likelihood gains $\Delta\text{MDL} > 0.01$ nats, while 15-topology `SPR2` quintet moves accept log-likelihood gains $\Delta \ln P > 0.05$ nats to prevent numerical oscillation. Under `-discrete` and `-ordered`, Stage 2 instead optimizes the exact discrete/ordered MDL objective directly via postorder/preorder Sankoff–Fitch/interval sweeps.

4. **Stage 3 — Deterministic Polytomy Contraction and Discrete Hub Expansion:** Internal branches contract sequentially from most negative gain upward when $G_{\text{MDL}}(e) \le 0$. On default continuous-time `GTR+CAT8`/`LG08+CAT8` paths, branch contraction tests $\Delta \ln P(e) - \left[\frac{1}{2}\ln\max(1, m_{\text{var}}/2\pi) + \frac{1}{2}\ln\max(1, N-3) + \mathcal{L}_{\text{topo}}(k_u, k_v)\right] \le 0$ and emits the contracted graph (alongside the pre-contraction binary refinement via `-b`) without running multi-taxon clique expansion; on `-discrete` and `-ordered` inputs, degree-d hubs additionally propose multi-taxon splits ($2 \le k_u \le d-2$) retained whenever $G_{\text{MDL}}(e) > 0$.

5. **Stage 4 — Dual-Mode Lateral Shunt Peeling:** Residual non-sister parallel substitutions $S_{u,v} = \{s_1, \ldots, s_m\}$ (pooled across up to three consecutive ancestral coding branches while excluding ancestral symplesiomorphies) are tested as lateral links $h = (u \to v)$ priced by endpoint address $2\ln|V|$ plus the shorter of contiguous-tract (`INTERVAL`) or scattered-site (`SPARSE`) mask codes:

$$\begin{aligned} \mathcal{L}_{\text{mask}}^{\text{INTERVAL}}([a,b]) &= \ln 2 + 2\ln L + \mathcal{L}_{\text{branch}}(d_{[a,b]}, b-a+1), \\ \mathcal{L}_{\text{mask}}^{\text{SPARSE}}(S_{u,v}) &= \ln 2 + \ln\binom{M_{\text{pool}}}{m}, \end{aligned} \tag{7}$$

   where $d_{[a,b]}$ counts discordant positions in $[a,b]$ and $M_{\text{pool}} \le L$ is the literal number of variable alignment columns. To guard against tearing genuine vertical tree splits on noisy tracts, Stage 4 masks recipient nucleotides only for contiguous `INTERVAL` transfers whose MDL gain exceeds 15 nats ($e^{-15} \approx 3 \times 10^{-7}$ false-masking odds) before running Stage 3 re-verification.

Multi-Alphabet Generalization and Continuous-Time Rate-Category Cache Tiling

Bushwhacker extends Eqs. (1)–(7) natively across discrete alphabets:

- **20-State Amino-Acid Alignments (`-protein` / `-translate`):** Twenty amino-acid bit-planes (`[u64; 20]` per 64 columns) and six Dayhoff–Miyata biochemical classes (*33*) yield empirical conservative probability $\hat{p}_{\text{cons}}$ (estimated from pairs with divergence ≤ 0.35) and normalized substitution costs:

$$\ell^{\text{AA}}_{\text{subst}}(a \to b) \;=\; \begin{cases} -\ln\!\big(\hat{p}_{\text{cons}}/k_{\text{same}}(a)\big), & \text{if class}(b) = \text{class}(a), \\ -\ln\!\big((1-\hat{p}_{\text{cons}})/k_{\text{diff}}(a)\big), & \text{if class}(b) \neq \text{class}(a). \end{cases} \tag{8}$$

- **Discrete Single-Cell Somatic Matrices (`-discrete`, $2 \leq K \leq 32$):** Binary somatic SNV profiles (0 = WT, 1 = mutant, ?/- = allelic dropout) assign full wildcard bitmasks ($2^K - 1$) to missing entries, incurring zero dropout penalty while adjusting $L_{\text{eff},i}$ per cell.

- **Ordered Zygosity and Copy-Number Profiles (`-ordered`, $3 \leq K \leq 16$):** Somatic zygosity states (0 = WT, 1 = HET, 2 = HOM/CN-LOH (*32*, *61*)) and DLP+ copy-number bins ($0, 1, \ldots, 11+$ (*62*)) replace unordered Fitch intersections with exact linear-time L_1 interval propagation: parent interval $[l_1, r_1] \cap [l_2, r_2]$ if overlapping, or $[\min(r_1, r_2), \max(l_1, l_2)]$ otherwise, paired with geometric step costs $\ell_{\text{step}}(\Delta s) = -\ln\left((1-\hat{q})\hat{q}^{|\Delta s|-1}/Z_a\right)$. On 500-kb bin matrices (4,375 bins), Bushwhacker supports raw bin compression with root run-length chromosomal coding and lossless breakpoint-segment aggregation (365–448 multiplicity-weighted segments).

- **Continuous-Time Hard-CAT Rate Models with 64-Site Cache Tiling (`GTR+CAT8` / `LG08+CAT8`):** Running unconditionally on nucleotide ($N \geq 4$) and protein/translated ($N \geq 3$) alignments (*16*–*18*), continuous-time reversible Markov models assign each variable column s unconditionally by nearest log-rate to one of eight fixed categories $(r_1, \ldots, r_8)$; analytical octile means of $\Gamma(1,1)$ satisfying $\frac{1}{8}\sum_{k=1}^{8} r_k = 1$):

$$\begin{aligned} G_{\text{CT-MDL}}(e) \;=\; & \Delta \ln P(e \mid T, r_{k^*(\cdot)}\hat{\mathbf{t}}, Q_M) \\ & - \left[\frac{1}{2}\ln \max\!\left(1, \frac{m_{\text{var}}}{2\pi}\right) + \frac{1}{2}\ln\max(1, N-3) + \mathcal{L}_{\text{topo}}(k_u, k_v)\right], \end{aligned} \tag{9}$$

  evaluated over cached transition matrices $P(t) = U\,\text{diag}(e^{\lambda r_k t})U^{-1}$ on a 15-point log-spaced branch-length grid with parabolic curvature refinement ($\frac{1}{2}\ln(1 + \mathcal{I}(e)/2\pi)$ during NNI scoring), contracting silent stems smoothly when $G_{\text{CT-MDL}}(e) \leq 0$. Partitioning unique site patterns into $B = 64$-pattern L1/L2 cache tiles bounds resident likelihood memory at $\mathcal{O}(N \cdot B \cdot C \cdot |\Sigma|)$ (0.95–2.39 GB vs. 26.4–79.2 GB for standard likelihood engines; Table S11). In JSON output, `tree_MDL_nats` records the unified discrete branch-edit summary code (Eq. (1) evaluated at continuous-time-informed branch loads $\max(\hat{t}_e \bar{r}_{\text{emp}} L_{\text{eff}}, m_e)$), whereas topological contraction on nucleotide and protein alignments evaluates the continuous-time criterion of Eq. (9) directly.

Lossless `.bwz` Causal Phylogenetic Arithmetic Codec

Invoked via `bushwhacker -c` and `-d`, the `.bwz` archiver encodes taxon identifiers, the Newick graph, the root sequence (with Markov run-length coding for indel and dropout runs), and causal preorder branch deltas using 32-bit arithmetic coding (*51*) conditioned on empirical site-rate and codon-phase priors, guaranteeing bit-exact FASTA reconstruction (`SHA-256` verified across Table S9).

Computational Environment, Baseline Software, and Multiple-Testing Governance

- **Implementation and Determinism:** Bushwhacker is written in native Rust (`rustc 1.94.1`) with zero external C/C++ dependencies and compiles to both a multi-threaded

CLI binary (`-T/-threads`, bit-exact identical across 1–64 threads with fixed seed `20261112` and zero random tie-breaking) and client-side WebAssembly (`bushwhacker_bg.wasm`). Open-source code, specimen files, reproducibility scripts, and browser UI reside at https://github.com/veg/bushwacker and https://veg.github.io/bushwacker/.

- **Compute Hardware:** All 32-core Tree-of-Life supermatrices, Greengenes 16S scaling runs, pandemic viral alignments, and coalescent cluster sweeps ran on an ARM64 Neoverse-N1 Linux server (64 cores, pinning 32 threads per method in head-to-head comparisons unless noted, with identical hardware wall-clock quotas and no timeouts bypassed). Single-cell cancer phylogenomics, specimen reconstructions, and `.bwz` archive verifications ran on an Apple M5 Max workstation (18-core CPU, 36 GB unified memory, macOS).

- **External Baseline Software:** Baselines were executed with `IQ-TREE multicore v2.4.0` (`-m MFP`, `-m GTR+I+G4`, `-m GTR+G4 -fast`, `-m LG+G4 -fast`, `-polytomy`, `-abayes`, `-alrt 1000`, `-B 1000`, and `AliSim`), `FastTree/FastTreeMP v2.1.11` (`-nt -gtr`, `-lg`), `ASTRAL-IV` (`ASTER astral4 -t 8`), `HyPhy GARD`, `INDELible`, `gzip -9`, `bzip2 v1.0.8 -9`, and `xz/lzma` (`XZ Utils v5.8.1`, `-6/-9`), scored by exact bipartition Robinson–Foulds matching (*72*).

- **Zero-Leakage, Multiple-Testing, and Ethics Governance:** Across every immune and cancer single-cell benchmark, cell multiplicities, DMS affinity scores, clinical timepoints, anatomic compartments, and passage numbers were stripped strictly prior to inference and linked solely post hoc. To control family-wise false discoveries across all 23 primary two-sided biological hypothesis tests reported across main text and supplement, raw p-values were adjusted using Benjamini–Hochberg (BH) step-up false discovery rate control ($\alpha_{\mathrm{FDR}} = 0.05$): 20/23 tests remain significant at $q < 0.05$, whereas the three null controls/non-significant comparisons remain explicitly null (pooled binary BCR hub-vs-nonhub DMS affinity $p = 0.203, q = 0.212$; Morita AML direct cell count vs. k_{out} $p = 0.197, q = 0.212$; and DLP+ `SA535` passage aneuploidy $p = 0.475, q = 0.475$). On the separate 20-state archaeal `ar122` Complexity Hypothesis screen, the nominal two-sided Fisher test gives $p = 0.0235$ ($q = 0.027$ alongside the same family). This retrospective *in silico* study analyzed only public de-identified datasets with zero wet-lab manipulation, human identifiers, or biosecurity concerns.

Data Provenance and Empirical Archive Identifiers

1. **Model-Selection Benchmark Archive (7,200 empirical + 27,500 simulated alignments):** OSF repository https://doi.org/10.17605/OSF.IO/T3PF2 (*14*, *35*), spanning PlantDB, Selectome, TreeBASE, and Pandora empirical loci alongside `INDELible` ground-truth suites `sim_bic` ($n = 5,901$), `c0` ($n = 7,199$), `c1` ($n = 7,200$), and `c2` ($n = 7,200$).

2. **Eukaryotic Phylogenomic Radiations (19 datasets, 45,205 loci; 83,180 multi-seed searches):** Figshare project `22040` (https://figshare.com/projects/Evaluating_fast_maximum_likelihood-based_phylogenetic_programs_using_empirical_phylogenomic_data_sets/22040 (*13*, *24*)), including mammalian `CHST13` (22 species, 1,176 bp from `TarvD7`), plus the whole-genome avian radiation (`Avian_Shultz2019`; Dryad https://doi.org/10.5061/dryad.4b8gt83 (*46*); 50 clean coding orthologs evaluated in `ASTRAL-IV` alongside 11 radiation benchmarks `ChenA4`, `MisoA2`, `ShenA9`, `BoroA6`, `WickA3/WickD3a`, `NagyA1`, `SongD1`, `XiD4`, `PrumD6`, `WhelA7` totaling 12,008 runs / 11,164 distinct loci).

3. **Mammalian *Prestin* (`SLC26A5`) and TOGA2 882-Mammal Atlas:** Complete 116-mammal *Prestin* coding alignment (2,235 bp; `docs/examples/prestin.fasta` (*22*, *23*)), and the Senckenberg TOGA2 `hg38 v1` archive (https://genome.senckenberg.de/

download/TOGA2/MultiCodonAlignments/reference_human_hg38/v1/ (*28*, *47*); 882 assemblies across 20 orders and 116 families: 19,890 non-degenerate coding loci, 19,952 translated proteins, 5,058 de novo `WALRUS` codon MSAs, and the 253-locus concatenated mammalian supermatrix).

4. **Botanical and Poaceae C_4 Corpus (2,002 alignments):** Land-plant `WickD3a`/`WickD3b` (1,688 alignments (*40*)), seed-plant `XiD4` (310 genes (*73*)), GenBank plastid `rbcL_483` ($483 \times 1,398$ bp), and Poaceae `ppc`, `pck`, and `rbcL` orthologs/paralogs including *Alloteropsis semialata* C_3/C_4 copies retrieved from NCBI Entrez Nucleotide (*49*, *74*, *75*).

5. **Prokaryotic Domains, Clinical *K. pneumoniae*, and Greengenes 16S:** GTDB Release 95.0 (https://data.gtdb.ecogenomic.org/releases/release95/95.0/ (*30*, *54*): `ar122` across 1,672 archaea and 1,000-species `bac120` across all 111 phyla); clinical *Klebsiella pneumoniae* genomes from NCBI BioProject `PRJNA246471` and RefSeq (*56*) (16-strain 214,038-bp core chromosome supermatrix, 222 single-copy core loci across 10 genomes, and matched 9 clinical isolates—`CAV1016`, `CAV1042`, `CAV1193`, `CAV1344`, `CAV1392`, `CAV1453`, `CAV1596`, `HS11286_ST11`, `NJST258_2`—pairing the 214,038-bp chromosome against their 882-bp *blaKPC* plasmid); *S. pneumoniae* `PMEN1` (11×2.22 Mb (*76*)); and Greengenes `13_5` 16S rRNA (https://greengenes.secondgenome.com/ (*58*), $N = 256 \rightarrow 32,768 \times 1,348$ bp).

6. **Pandemic Viral and `.bwz` Archives:** Open Nextstrain/GenBank SARS-CoV-2 (https://data.nextstrain.org/files/ncov/open/; 5,097 and 10,062 complete 29,903-bp genomes plus blinded Pango `X*` recombinants), global Dengue `DENV-1..4` (1,426, 2,632, 855, and 585 coding genomes; 5,498 total), LANL HIV-1 (https://www.hiv.lanl.gov/; `KAL153`, `CRF02_AG`, `HIV1_518` (*50*, *77*)), bat sarbecoviruses (`Sarbeco_83`, $83 \times 30,965$ bp (*52*, *53*)), `InfluenzaA` HA (349×987 bp), `Human_mtDNA` ($78 \times 16,585$ bp), hominoid `MHC` ($379 \times 1,128$ bp (*21*)), and *Heliconius* `optix` (*78*, *79*).

7. **Single-Cell Immune and Cancer Cohorts:** Mouse `CGG`-immunized germinal-center repertoires from `GCReplay` (https://github.com/matsengrp/gcreplay (*31*); 147 GCs, 10,429 paired `IGH+IGK` cells, 9,352 unique 657-bp genotypes, plus 4,400 scFv yeast-display $\Delta \log_{10} K_D$ scores) and `Clone_3-8` (66 cells (*59*)); Mission Bio Tapestri myeloid scDNA runs from `COMPASS` (Zenodo https://doi.org/10.5281/zenodo.10822292 (*61*); 168 runs, 1,001,609 cells) and the 123-patient AML cohort including specimen `AML-38-001` (7,016 cells, 11 clones $\times 31$ loci; NCBI GEO `GSE146150` and Zenodo https://zenodo.org/records/15096570 (*32*)); and DLP+ scWGS copy-number matrices (https://doi.org/10.24072/pcjournal.292 (*62*); 4,950 cells across `OV2295`, `SA501`, `SA609`, `SA535`).

8. **Ground-Truth Synthetic Benchmarks (36,920 + 50 GC histories):** `AliSim` 22-regime nucleotide suite (220 alignments (*34*)), `Abadi/Gerth` nucleotide suite (27,500 alignments), `LG+G4` protein suite (`P1-P4`, 400 alignments), deterministic Kingman Anomaly-Zone coalescent suite (`seed = 20261112`) (8,000 loci (*36*)) + 50 Galton–Watson B-cell GCs (`MK_RS5NF` 5-mer SHM (*60*)), reticulate HGT/convergence suite (`N1-N4`, 400 alignments), and single-cell SNV/CNA matrices (`C1-C4`, 400 matrices).

Supplementary Text

Section S1: Why Two-Part Compression Resists Long-Branch Attraction and Enforces Polytomies

Counting edits along branches superficially resembles unweighted maximum parsimony (*1*), which falls prey to long-branch attraction under rate heterogeneity (*15*). Eqs. (1)–(9) prevent this artifact in three ways. First, Eq. (4) weights every shifted mutation by $w_c = \ln \frac{L_{\text{eff}} - m_c}{m_c + 1}$: on our

worked 4-taxon, 16-site quartet (88.72 nats raw, 22.18 nats root, 61.02 nats star at $m_e = 0$ with $m_c = 2$ private leaf edits costing $\mathcal{L}_{\text{branch}}(2, 16) = 9.71$ nats each; Fig. S1 and Table S5), pooling a shared change from two quiet branches ($m_{\text{short}} = 1$, $w_{\text{short}} = 2.01$ nats) saves 4.03 nats versus 0.71 nats on two six-mutation branches ($w_{\text{long}} = 0.36$ nats)—a 5.6-fold positional shield matching Felsenstein's likelihood weights (*2*). Second, with $m_c = 2$ private edits per leaf ($\mathcal{L}_{\text{branch}}(1..4, 16) = 5.56, 9.71, 12.99, 15.72$ nats), one shared mutation saves 6.55 nats on the daughter branches yet costs $\mathcal{L}_{\text{branch}}(1, 16) + \ln 3 = 6.66$ nats, contracting automatically ($G_{\text{MDL}} = -0.11$ nats, $P_{\text{local}} = 0.473$), whereas $m_e = 2$ (+12.02 vs. 10.81, $G_{\text{MDL}} = +1.21$ nats, $P_{\text{local}} = 0.771$) and $m_e = 3$ (+16.69 vs. 14.08, $G_{\text{MDL}} = +2.60$ nats, $P_{\text{local}} = 0.931$; 75.11 nats resolved total) pay for the branch header. Third, continuous-time hard-CAT `GTR+CAT8`/`LG08+CAT8` pruning (Eq. (9)) handles multi-hit saturation directly across nucleotide and protein alignments.

Section S2: Synthetic Validation (36,970 Histories) and Model-Selection Reconciliation

Across all simulated regimes (Figs. S2 and S3, Tables S2–S5), pricing split addresses eliminates overfitting without losing power as data accumulate. On 128-taxon star bursts (`S08`), Bushwhacker emits 0.0 false splits (versus 125.0 for binary ML, 74.8 for `IQ-TREE -polytomy`, 11.9 for `FastTree SH90`, and 2.3 for `UFBoot95`), rises from 8.2% recall (16.3/197, 99.7% precision at $L = 100$ bp) to 99.5% recall (196.0/197, 100.0% precision at $L = 16{,}000$ bp), recovers 96.0% of Felsenstein-zone splits (0% LBA), maintains 99.2% sister-clade recall across $m = 0 \to 5$ convergent codons, cuts BioNJ error on 256-taxon pectinate trees, and improves out-of-sample held-out compression by +64.3 nats on multifurcations (RF = 0.38 vs. 16.0) and +400.9 nats under $m = 3$ convergence over binary ML (which exhibits +3.90 and +126.3 nats of in-sample likelihood excess). Across 44,338 tested splits, conditional local posteriors $P_{\text{local}}(e \mid T_{-e}, A)$ are well-calibrated at $P_{\text{local}} \geq 99\%$ (89.4% of splits, 98.3% true; overall ECE = 3.74%) while remaining optimistic on intermediate $[0.50, 0.99)$ splits (40.6%–80.3% true; Table S3) because P_{local} conditions on a single optimized topology T_{-e}.

On the 7,200 empirical and 27,500 matched simulated alignments of Abadi et al. (*14*) and Gerth (*35*) (Table S4), 24,338 splits flip between BIC ModelFinder and GTR + I + G4, and 74.7% carry < 1.0 expected substitution; Bushwhacker contracts 96.4% (23,465) under deterministic heuristic search (baseline IQ-TREE model comparisons used default time-seeded runs with `-fast` for $N > 35$ and on `c0-c2`). This reconciles both studies (*80*, *81*): average binary RF is nearly identical across models ($\text{RF}_{\text{bin}} = 8.42$ vs. 8.46 on `sim_bic`), yet on radiation-rich trees (> 50% short branches, $n = 625$), binary ML invents 22.62 false splits/tree (33.8% precision) and `-polytomy` invents 11.79 (50.2%), compared with 0.18 (98.8% precision) under Bushwhacker. Auditing Abadi et al. (*14*)'s `phyml -o lr` reference protocol further confirmed that identical alignments produce zero-length branches resolved identically by alphabetical BioNJ tie-breaking across models.

Section S3: Eukaryotic Radiations, Coalescent Species Trees, and Plant C_4 Networks

Across six multi-seed eukaryotic benchmarks (4,159 gene-alignments across 3,315 unique genes, 83,180 independent RAxML/IQ-TREE searches (*24*); Table S6), 48.2% of binary splits disagree across searches ("search-unstable splits"); Bushwhacker contracts 94.5% (155,821/164,901) while $\rho(A)$ quartiles track search stability (60.2% → 35.2%). Across all 19 eukaryotic datasets (45,205 loci (*13*, *82*); exemplified by mammalian *CHST13* removing $8 \to 0$ conflicts while retaining 9 clades), withholding $G_{\text{MDL}} \leq 0$ splits eliminates 61.5% (936,166/1,522,139) of apparent conflicts and resolves contested nodes via cumulative $\Delta\mathcal{L}_{\text{MDL}}$ (`BoroA6` Ctenophora 38.7 vs. 5.5 nats; `WhelA7` 248.7 vs. 84.5; `ChenA4` Lungfish 6,976.8 vs. 5,781.2; Archelosauria 16,235.1 vs. 1,596.0; Atlantogenata 2,364.8 vs. 769.6).

Passing multifurcating gene trees directly into summary coalescent solvers (`ASTRAL-III`/`ASTRAL-IV` (*37*, *83*)) improves coalescent species-tree inference without bootstrap cutoffs in Anomaly-Zone simulations (nRF = 0.2048 vs. 0.2429–0.2762 for `FastTree` across 2,000 loci; Table S2) and across empirical radiations spanning jawed vertebrates (`ChenA4`: nRF = 0.0727 vs. 0.1091 binary), Neoaves birds (`PrumD6`: 0.1624 vs. 0.1726), fungi (`NagyA1`: 0.1754 vs. 0.1930), whole-genome birds (Shultz and Sackton (*46*): 0.2778 vs. 0.3056), and placental mammals (`SongD1`: nRF = 0.0294 matching `IQ-TREE-10`), while exposing the boundary on ultra-short (L = 300 bp) or Precambrian single-gene markers where contracting 70–75% of splits removes weak single-substitution signal. In land plants (`WickD3a` → `WickD3b`, 844 loci; Fig. S5D), dropping third codon positions eliminates 88.3% (60 → 7) of GC/AT `SPARSE` artifacts while preserving *Sorghum* ↔ *Zea* conversion (+8.04 nats); in Poaceae, Bushwhacker recovers native C_3 *Alloteropsis* paralogs inside Paniceae (P_{local} = 1.000), places horizontally transferred C_4 `pck-1P1`/`ppc-1P3` inside donor clades (P_{local} = 1.000), and isolates post-transfer catalytic convergence (+19.66 nats across 27 `ppc` codons; +17.58 nats across 12 `pck` codons).

Section S4: Mammalian *Prestin* and Proteome-Wide TOGA2 Census (19,952 Loci)

On the 116-mammal *Prestin* (*SLC26A5*, 2,235 bp) coding alignment (*22*, *23*) (Fig. S4), Bushwhacker resolves 81/113 binary splits (19 polytomy hubs), maintains monophyletic Cetacea within Artiodactyla (P_{local} = 1.00) and monophyletic Chiroptera (P_{local} = 1.00; 76 TP, 5 FP, 93.8% precision, nRF = 0.2165 in 0.055–0.129 s vs. 18 FP/83.6% `IQ-TREE -polytomy`, 3 FP/96.5% `UFBoot95` [73.5% rec.], 4 FP/95.2% `SH-aLRT80` [69.9% rec.], 19 FP/83.2% raw `IQ-TREE 2` [78.9 s, 0 shunts], and 18 FP/84.1% `FastTree 2`), while certifying two `SPARSE` lateral links: a primary echolocation shunt (+6.64 nats, P_{local} = 99.87%) uniting Delphinidae and Rhinolophoidea bats across the four classic motor-domain codons (`N7T`, `I384T`, `S392A`, `G601A`), and a secondary link between *Rhinolophus sinicus* and toothed whales (+2.79 nats, P_{local} = 94.22%, codons `308, 554, 566, 621, 685`); canonical site `L497M` is shared by non-echolocating *Pteropus* fruit bats and rightly filtered as an ancestral Chiroptera state.

Across 19,952 TOGA2 mammalian orthologs up to 882 species (*28*, *47*) (Fig. S5A–C and Table S7), coding contraction removes 55.5% (3.87 million) of species-tree clashes down to a length-invariant ∼ 33.2–34.0% lineage-sorting/alignment floor (trimmed 21.2% on 5,058 de novo `WALRUS` codon MSAs), while 20-state proteins eliminate 72.7% (7.87 million) of conflicts and contract 65.0% overall (5,226,678/14,937,057 binary splits retained; 2,952,245 conflicting). Spatial pricing separates intra-family tandem block conversion (4,202 AA tracts, median 144 bp, 81.5% within families / 94.2% intra-order, 3.4–4.5× enriched in keratins, cadherins, `TRIM`, KRAB-ZNF, and globins vs. 3.1% single-copy) from cross-order convergence (11,713 `SPARSE` events, 45.8% cross-order, 7.9× enriched), certifying 16/25 (64%) core cochlear hair-cell genes (*84*, *85*) (`PCDH15`: +54.6 nats max single AA shunt, +93.2 nats all-mode cumulative, +30.7 nats species-tree AA `SPARSE`; +4.0 nats on 875-species *SLC26A5* AA) alongside neutral proteome-wide bat–whale counts (0.43×) versus strong genome-wide excess in ant-eating (25.8× raw $p = 1.3 \times 10^{-19}$; 18.6× family-deduplicated across 13 independent mammalian families, $p = 8.1 \times 10^{-13}, q = 3.7 \times 10^{-12}$) and subterranean blind mammals (4.0× across 6 families, $p = 0.0045, q = 0.0057$).

Section S5: Prokaryotic Domains, Clinical Plasmids, 16S Scaling, and `.bwz` Mosaicism

Across 242 GTDB r95 single-gene markers (*30*, *54*) (Table S8), contracting saturated deep splits cuts 42.5% (61.1% in Bacteria) of discordant branches, raises bacterial genus compatibility to 92.3% (vs. 86.0% FastTree 2), and in `ar122` 20-state AA space recovers 20.6% (21/102) operational metabolic transfers (including DPANN Nanoarchaeota exchanges (*57*)) versus 0.0% (0/20, nominal two-sided Fisher $p = 0.0235, q = 0.027$) across core ribosomal/RNAP subunits

(*29*, *55*). In hospital *Klebsiella pneumoniae* (*56*), the 214-kb core chromosome resolves 6/6 splits ($P_{\text{local}} = 1.00$, 0.5% lateral budget; 13/13 splits across the 16-strain supermatrix, supplemented by 222 core loci across 10 genomes), whereas the matched 882-bp *blaKPC* plasmid collapses 5/6 splits (83.3%) into *blaKPC-2* and *blaKPC-3* (`C814T`) horizontal hubs. Scaling Greengenes 16S rRNA across eight nested tiers ($N = 256 \rightarrow 32,768$ OTUs (*58*)) reduces supported splits from 88.5% ($N = 256$) and 91.4% ($N = 1,024$) to 76.0% ($N = 32,768$; 4,585 hubs), cuts contradicting splits per taxon group by 19.4–23.8% across five ranks relative to forced-binary trees (while forced-binary `FastTreeMP` retains higher strict monophyly, 61.4–75.5% vs. 32.0–44.2%), runs 4.50× faster than 32-core `FastTreeMP` (251.6 s vs. 1,133.2 s at $N = 32,768$), and archives data at 64.43 B/OTU (2,111,286 B, 1.08× smaller than `xz -6`). Across Table S9, `.bwz` sequence payloads outperform general dictionary compressors by 1.01 × –2.06× (1.6 × –18.9× `gzip -9`; full archives 63.2 B/genome [1.25 × xz, 11.5 × gzip] on 10,062 SARS-CoV-2) across clonal viral, mitochondrial, plastid, and 2.22-Mb *S. pneumoniae* `PMEN1` (*76*, *86*) alignments until > 1–4 whole-genome breakpoints (`Sarbeco_83` 0.71× payload [0.68× archive], `HIV1_CRF02_AG` 0.65× payload) let multi-donor `lzma -9` dictionaries (*87*) overtake single-tree coding.

Section S6: Germinal-Center B Cells and Single-Cell Human Cancer Cohorts

On 50 Galton–Watson affinity-maturation GCs (90 sampled cells plus germline under 5-mer SHM (*60*); Table S2), zero-leakage Bushwhacker achieves 99.83% unique-genotype split precision (0.02 vs. 42.34 FP/GC for binary ML; 0.10 s/GC; cell-level $F_1 = 0.7533, 0.12$ FP) and recovers mutant burst degree ($\rho = 0.438, p = 2.9 \times 10^{-181}$). Across all 147 mouse lymph-node GCs (10,429 B cells, 9,352 unique 657-bp genotypes (*31*)) and `Clone_3-8` (66 cells (*59*)), MDL eliminates binary staircase errors to 0.00 FP/GC at cell level (93.2% precision, 57.3 s vs. 1,807.1 s `IQ-TREE 2`) and 0.007 FP/GC on unique genotypes (95.1% precision, 54.3 s vs. 1,524.6 s; 1/63 vs. 63/63 splits on `Clone_3-8`). Post hoc joining shows monotonic expansion from $k = 0$ leaves (5.1% expanded ≥ 2 cells) to $k \geq 3$ hubs (48.2%; 9.5×, paired GC Wilcoxon $n = 86, p = 5.9 \times 10^{-6}, q = 1.3 \times 10^{-5}$; mouse $n = 25, p = 2.0 \times 10^{-6}, q = 6.5 \times 10^{-6}$), continuous within-GC correlation with yeast-display $\Delta \log_{10} K_D$ affinity (GC $n = 127$, Spearman Wilcoxon $p = 1.7 \times 10^{-3}, q = 2.5 \times 10^{-3}$; mouse $n = 28, p = 2.9 \times 10^{-3}$, whereas coarse binary hub-vs-nonhub pooling is null at $p = 0.203, q = 0.212$), and 5.6× odds enrichment in CDRs among mutations recurring across ≥ 5 mice (64.0% [16/25] vs. 24.1% [61/253], OR $= 5.60, p = 7.5 \times 10^{-5}, q = 1.6 \times 10^{-4}$; DMS +0.305 vs. $-0.083, p = 8.1 \times 10^{-5}$).

In human tumors with cell counts and labels withheld (Fig. S6 and Table S10), ordered 3-state compression across 168 COMPASS myeloid Tapestri runs (1,001,609 cells, 6,222 genotypes (*61*); 2.62 s) recovers 6.18× larger in vivo clone sizes at $k \geq 3$ hubs (232.8 vs. 37.7 cells; per-run Wilcoxon $n = 166, p = 1.7 \times 10^{-18}, q = 1.9 \times 10^{-17}$; per-patient $n = 127, p = 3.5 \times 10^{-14}$; Spearman $\rho = 0.24$) and 90.3% (2,858/3,164) stepwise WT → HET → HOM CN-LOH transitions; across the independent 123-patient Morita AML cohort (0.128 s (*32*)), branching hubs produce larger descendant lineages ($n = 97$, paired Wilcoxon $p = 3.0 \times 10^{-15}$, whereas direct hub cell multiplicity is null at $p = 0.197$), late subclonal signaling resistance drivers enrich relative to epigenetic founders (OR $= 1.79, p = 1.9 \times 10^{-3}, q = 2.7 \times 10^{-3}$; parallel drivers in 44 patients), and specimen `AML-38-001` (3.0 ms) resolves an *NPM1* trunk branching into parallel *IDH1* ($P_{\text{local}} = 0.996, k = 3$) and *IDH2* ($P_{\text{local}} = 0.997, k = 4$) signaling hubs. Across 4,950 DLP+ scWGS copy-number genomes (4,375 500-kb bins, 12 states (*62*)), ordered step coding finishes in 21.9 s on colinear segments (113.6 s raw bins; 21.9–33.5× raw-bin compression), contracts 33.1–49.9% (39.0–53.1% segments) of noisy binary splits while preserving cohort clade purity (57.9%–95.7% seg / 60.7%–95.7% bin vs. 20.8%–59.9% random null and 60.1%–96.7% forced-binary), isolates primary ovary vs. ascites subclones (`OV2295` 95.7% clade purity, weighted $F_1 = 0.997$ seg / 0.995 bin), and notes sample-level HGSOC aneuploidy rank in `OV2295` ($\rho = 0.818, q = 1.1 \times 10^{-266}$, an input property; TNBC PDX `SA609`/`SA501`/`SA535` $\rho = -0.11, -0.05, +0.03$, $p = 0.475$ on `SA535`).

Supplementary Figures

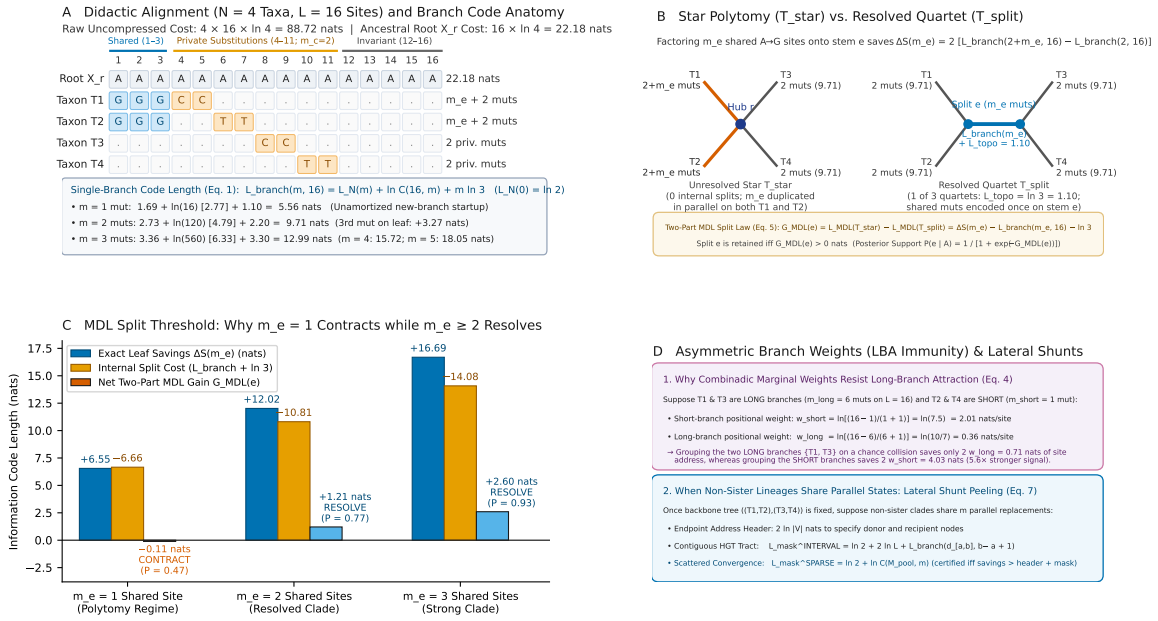


Figure S1: **Worked two-part MDL ledger on a 4-taxon, 16-site alignment ($N = 4, L = 16$).** (**A**) Alignment matrix ($X_r = \mathtt{A}^{16}$, 88.72 nats raw, 22.18 nats root) and single-branch code lengths $\mathcal{L}_{\text{branch}}(m, 16) = 5.56, 9.71, 12.99$ nats ($m = 1, 2, 3$) from Eq. (1). (**B**) Unresolved star baseline T_{star} (61.02 nats) versus resolved quartet ($\mathcal{L}_{\text{topo}}(2, 2) = \ln 3 = 1.10$ nats; 75.11 nats total at $m_e = 3$). (**C**) Exact leaf savings $\Delta\mathcal{L}_{\text{children}}(m_e)$, split cost, and net gain $G_{\text{MDL}}(e)$ for $m_e = 1$ (-0.11 nats, $P_{\text{local}} = 0.47$, contracted), $m_e = 2$ ($+1.21$ nats, $P_{\text{local}} = 0.77$), and $m_e = 3$ ($+2.60$ nats, $P_{\text{local}} = 0.93$). (**D**) Combinadic positional asymmetry ($w_{\text{short}} = 2.01$ vs. $w_{\text{long}} = 0.36$ nats) and Eq. (7) shunt pricing.

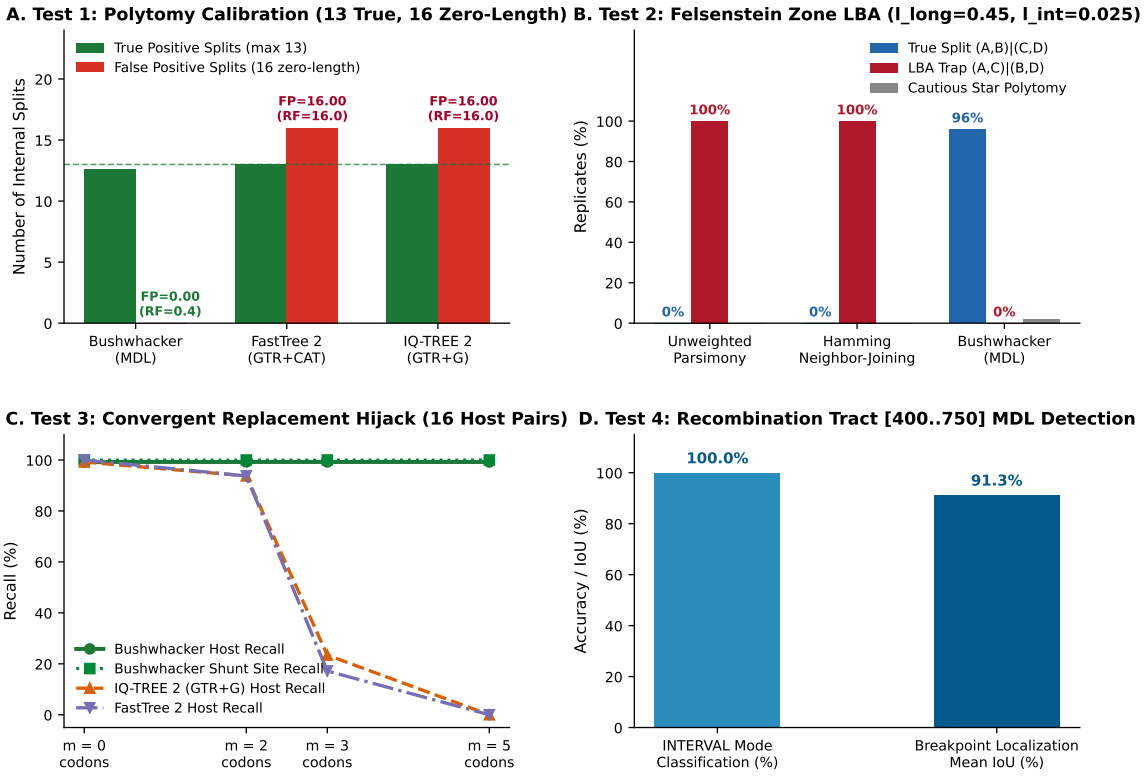


Figure S2: **Ground-truth control benchmarks across polytomy calibration, Felsenstein long-branch attraction, convergent hijack, and recombination tracts.** (**A**) Polytomy test (13 true and 16 zero-length internal splits): Bushwhacker emits 0.00 false splits (RF = 0.4) versus 16.00 false splits under forced-binary `FastTree 2` and `IQ-TREE 2`. (**B**) Felsenstein Zone ($l_{\text{long}} = 0.45$, $l_{\text{int}} = 0.025$): 96% true split recovery and 0% long-branch attraction versus 100% attraction under unweighted parsimony and neighbor-joining. (**C**) Sister-clade preservation across $m = 0 \rightarrow 5$ convergent codons (16 host pairs; 99.2% Bushwhacker recall versus 0% binary ML recall at $m = 5$). (**D**) Exact `INTERVAL` classification (100.0%) and breakpoint localization (IoU = 91.3%) on $[400, 750]$ mosaic tracts.

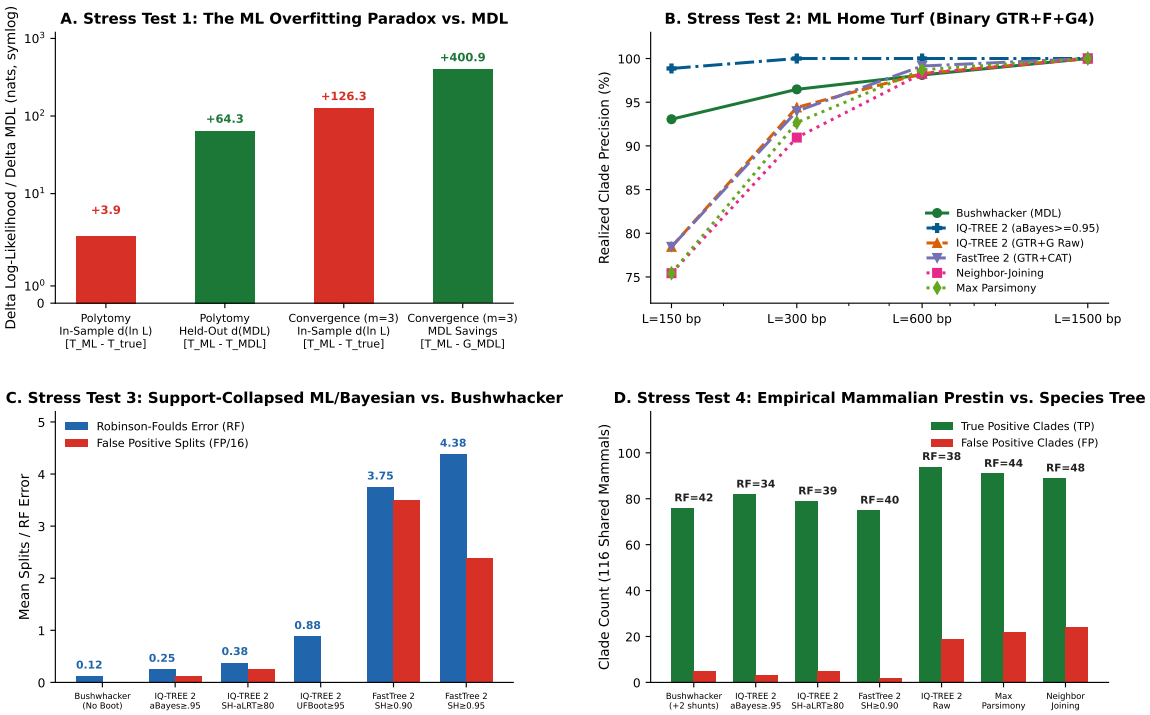


Figure S3: **Overfitting controls, support-collapsed baselines, and empirical mammalian *Prestin* comparison.** (**A**) In-sample binary ML likelihood excess (+3.90 and +126.3 nats) versus out-of-sample held-out compression advantage (+64.3 nats on multifurcations; +400.9 nats under $m = 3$ convergence). (**B**) Realized clade precision on strictly binary `GTR+F+G4` trees across $L = 150 \rightarrow 1,500$ bp. (**C**) Mean RF error and false splits for Bushwhacker (0.12 RF) versus `IQ-TREE 2` (`aBayes`, `SH-aLRT`, `UFBoot`) and `FastTree SH` cutoffs. (**D**) Empirical 116-mammal *Prestin* split recovery relative to the NCBI mammalian species tree (all 116 taxa; 76 TP, 5 FP, 93.8% precision under Bushwhacker with +2 certified echolocation shunts).

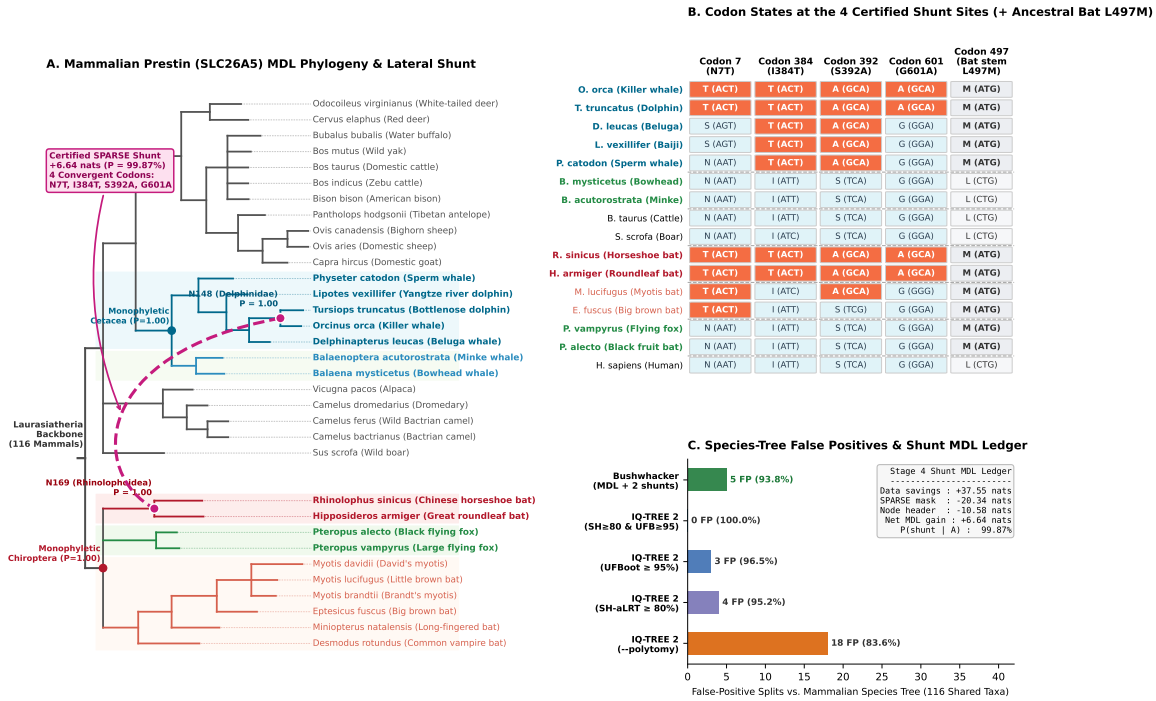


Figure S4: **Unsupervised graph induction on 116-mammal *Prestin* (SLC26A5) preserves cetacean and bat monophyly while isolating convergent echolocation codons.** (**A**) Laurasiatheria subtree showing monophyletic Cetacea within Artiodactyla ($P_{\text{local}} = 1.00$), monophyletic Chiroptera ($P_{\text{local}} = 1.00$), and the primary certified `SPARSE` lateral link ($+6.64$ nats, $P_{\text{local}} = 99.87\%$) uniting Delphinidae and Rhinolophoidea bats (alongside a secondary $+2.79$ nats link, $P_{\text{local}} = 94.22\%$, connecting *Rhinolophus sinicus* and toothed whales across codons `308, 554, 566, 621, 685`). (**B**) Codon and amino-acid states across the four classic echolocation sites (`N7T`, `I384T`, `S392A`, `G601A`), contrasted with Chiroptera stem substitution `L497M` shared by non-echolocating *Pteropus* fruit bats. (**C**) False-positive species-tree splits across inference methods (5 FP, 93.8% precision for Bushwhacker vs. 3 FP/96.5% `IQ-TREE UFBoot95`, 4 FP/95.2% `SH-aLRT80`, 18 FP/83.6% `-polytomy`, and 19 FP/83.2% raw `IQ-TREE 2` [78.9 s]) and exact two-part MDL shunt ledger.

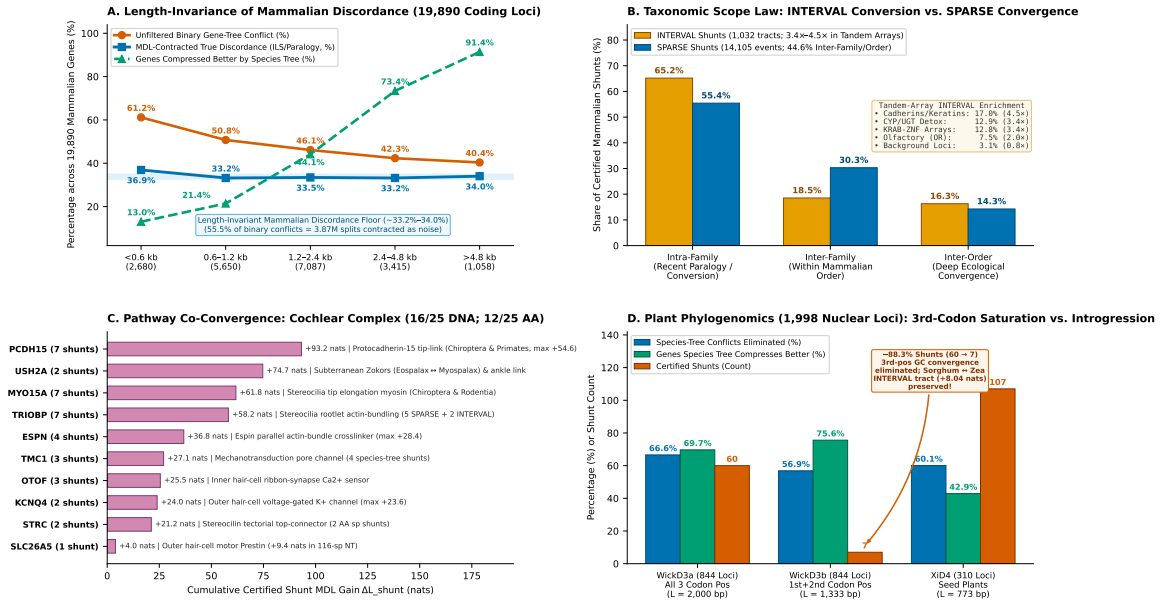


Figure S5: **Mammalian coding-DNA discordance floor, paralog conversion versus cross-order convergence, cochlear co-convergence, and land-plant saturation.** (**A**) Length stratification across 19,890 TOGA2 coding genes showing 55.5% (3.87 million) conflict removal down to a $\sim$ 33.2–34.0% genealogical floor. (**B**) Taxonomic scope of contiguous block conversion (1,032 tracts; 3.4–4.5$\times$ tandem-array enrichment) versus scattered parallel links (14,105 events). (**C**) Cumulative lateral shunt gains across top cochlear hair-cell genes (`PCDH15` +93.2 nats, `USH2A` +74.7 nats, `MYO15A` +61.8 nats; 875-species AA *SLC26A5* +4.0 nats, 116-species coding *Prestin* +9.4 nats cumulative). (**D**) Land-plant all-codon (`WickD3a`) versus codons $1+2$ (`WickD3b`) eliminating 88.3% ($60 \rightarrow 7$) of GC/AT artifacts while keeping *Sorghum*–*Zea* conversion (+8.04 nats).

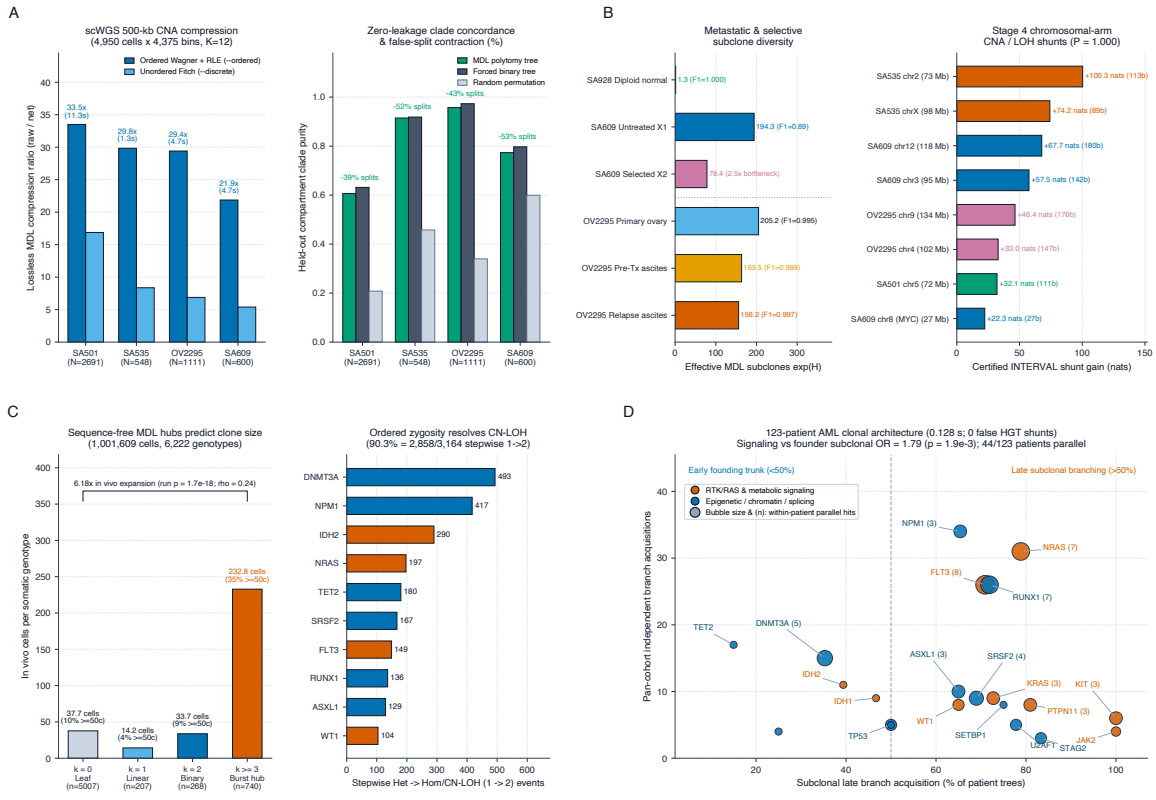


Figure S6: **Zero-leakage single-cell cancer phylogenomics across 4,950 DLP+ whole-genome copy-number profiles and 1,001,609 myeloid cells.** (**A**) Ordered 12-state copy-number compression (21.9–33.5× raw bins across `SA501`, `SA535`, `OV2295`, `SA609`) alongside held-out anatomic/passage clade purity (`OV2295` 95.7% purity, $F_1 = 0.997$ seg / 0.995 bin) and 33.1–49.9% (39.0–53.1% segments) false-split contraction. (**B**) Subclonal diversity bottlenecks and certified chromosomal-arm CNA/LOH tracts including `MYC` and `TP53`. (**C**) Across 168 Tapestri myeloid runs (1,001,609 cells, 6,222 clones (*61*)), inferred hub out-degree $k \geq 3$ predicts 6.18× larger in vivo clone sizes (per-run Wilcoxon $p = 1.7 \times 10^{-18}, q = 1.9 \times 10^{-17}$; $\rho = 0.24$) and captures 90.3% (2,858/3,164) stepwise WT → HET → HOM CN-LOH transitions. (**D**) Trunk-versus-subclone driver landscape across 123 AML patients (*32*) (OR = 1.79 late signaling enrichment, $p = 1.9 \times 10^{-3}, q = 2.7 \times 10^{-3}$; parallel acquisitions in 44 patients).

Supplementary Tables

Table S1: **Master overview of empirical biological corpora and ground-truth simulation suites evaluated with Bushwhacker.** All single-cell abundance counts, DMS binding affinities, clinical timepoints, and taxonomic labels were strictly withheld during tree and network inference.

Biological Scale / Benchmark Suite	Alphabet / Mode	Datasets / Runs	Taxa or Cells ($\times L$)	Primary Biological / Statistical Target	Key Finding
Model-Selection Archive (*14*, *35*)	Nucleotide (4-state)	7,200	7–150 × 100–8,000 bp	BIC vs. GTR+I+G4 vs. JC discordance	96.4% of model-conflicting splits contracted
Eukaryotic Radiations (*13*, *24*, *46*)	NT & 20-state AA	45,205 + 12 `ASTER`	19 clades (83,180 ML runs)	Gene-tree conflict & `ASTRAL-IV`	94.5% search-unstable & 61.5% sp. conflict cut
Mammalian Proteome (TOGA2 (*28*))	Codon NT & 20-state AA	39,904 (19,952 × 2)	Up to 882 sp. (20 orders)	Species concordance, conversion, convergence	72.7% AA conflict cut; 18.6× fam. / 25.8× raw ant-eater conv.
Mammalian Supermatrix (*28*)	GTR+CAT8 & LG08+CAT8	2 (253 loci)	882 × 543,864 bp/181,288 AA	Deep mammalian Tree of Life	92.7% IQ-TREE agreement; 24–37× faster (193–288× FT)
Prokaryotic Domains (GTDB r95 (*30*))	NT, AA, & LG08+CAT8	434 + 2 supermatrices	1,672 Archaea; 1,000 Bacteria	Precambrian phyla & Complexity Hypothesis	82.4–94.3% phyla; 20.6% vs. 0% HGT ($q = 0.027$)
Clinical *K. pneumoniae* (*56*)	NT & 20-state AA	9–16 str.; 222 loci	9–16 genomes (214 kb + *blaKPC*)	Chromosomal core vs. resistance plasmid	99.5% vertical core vs. 83.3% star plasmid
Botanical & Poaceae C_4 (*40*, *49*)	Codon NT & 20-state AA	2,002	Up to 483 plant species	3rd-codon GC/AT bias & C_4 HGT + conv.	88.3% 3rd-pos. shunts removed; C_4 HGT+conv.
Pandemic Viruses (SARS-CoV-2, DENV, HIV)	NT, AA, & `.bwz`	16 cohorts	10,062 SC2; 5,498 DENV-1..4	Superspreading hubs, recombinants, `.bwz`	38.5 B payload / 63.2 B arch. (18.9×/11.5× gzip); 76,000× GARD
Microbiome 16S rRNA (*58*)	Nucleotide & `.bwz`	8 scaling tiers	256 → 32,768 × 1,348 bp	Single-marker topological saturation	91.4% → 76.0% supported; 4.5× FastTreeMP
Germinal-Center B Cells (*31*, *59*)	Nucleotide (deduplicated)	148 GCs (147 + 1)	10,495 B cells (9,394 geno.)	In vivo clonal bursts & DMS K_D affinity	9.5× expansion ($p = 5.9 \times 10^{-6}$); 5.6× CDR odds
Human Myeloid scDNA (*32*, *61*)	Ordered 3-state zygosity	291 (168 runs + 123 pts)	1,001,609 cells (6,222 clones)	Leukemic expansion, CN-LOH, late signaling	6.18× hub cells ($p = 1.7 \times 10^{-18}$); 90.3% LOH
Human Tumor DLP+ scWGS (*62*)	Ordered 12-state CNA	4 cohorts (seg + bin)	4,950 cells ×4,375 500-kb bins	Subclonal aneuploidy & ascites evolution	21.9 s seg (21.9–33.5× comp.); $F_1 = 0.997$ purity
Ground-Truth Simulations (*14*, *34*)	NT, AA, MSC, Net, Somatic	36,920 (+ 50 GCs)	7–1,024 taxa; $L = 40$–16,000	Polytomies, calibration, HGT, single-cell	99.7–100% burst prec.; 3.74% ECE; 0% HGT FPR

Table S2: **Multi-alphabet ground-truth validation across 20-state proteins, Anomaly-Zone multi-species coalescence, reticulate networks, and single-cell somatic profiles (9,250 histories; complementing the 27,720 nucleotide alignments in Tables S4 and S5).**

Suite & Ground-Truth Regime	Method / Configuration	Res / True	FP / Tree	Precision	Recall	nRF / Auxiliary Metric	Time (s)
Protein Suite (`LG+G4`, $N = 64, L = 400$ AA)							
`P1` Star Burst (100 reps, 0 true splits)	Bushwhacker (`-protein` MDL)	0.05/0.0	0.05	96.0% (0-FP)	—	0.08% FPR	0.81
	IQ-TREE 2 (`LG+G4 -polytomy`)	38.67/0.0	38.67	0.0%	—	63.4% FPR	9.08
	IQ-TREE 2 / FastTree 2 (Binary)	61.00/0.0	61.00	0.0%	—	100.0% FPR	9.11/4.59
`P2` Rapid Radiation (50% silent stems)	Bushwhacker (`-protein` MDL)	30.14/30.49	0.05	99.8%	98.6%	nRF = 0.0077	1.39
	IQ-TREE 2 (`-polytomy`)	50.31/30.49	19.84	60.5%	99.9%	nRF = 0.3362	8.06
	IQ-TREE 2 / FastTree 2 (Binary)	61.00/30.49	30.55/30.54	49.9%	99.9%	nRF = 0.5189	8.01/3.84
`P3` Standard Resolved Yule (61 splits)	Bushwhacker (`-protein` MDL)	59.60/61.0	0.01	100.0%	97.7%	nRF = 0.0116	0.99
	IQ-TREE 2 (`LG+G4`)	61.00/61.0	0.08	99.9%	99.9%	nRF = 0.0013	7.04
`P4` Convergent Protein Sites (45 splits)	Bushwhacker (`-protein` + Stage 4)	44.40/45.0	0.01	100.0%	98.6%	92.8% conv. AA site prec.; 0% false clade	0.89
Coalescent MSC Suite (8,000 loci → `ASTRAL-IV`)							
24-sp. Asymmetric Anomaly Zone ($L = 400$ bp)	Bushwhacker (MDL) → ASTRAL-IV	37.8% gene res.	—	—	—	Species nRF = 0.2048 (−15.6% vs. FT)	0.47
	FastTree 2 (raw / SH-collapsed) → ASTRAL	100% / SH	—	—	—	Species nRF = 0.2429–0.2762	2.10
32-sp. Rapid Radiation ($L = 800$ bp)	Bushwhacker (MDL) → ASTRAL-IV	54.1% gene res.	—	—	—	Species nRF = 0.0586 (ties IQ `-poly`)	0.62
50-Galton–Watson B-Cell GCs (90 cells/GC)	Bushwhacker (MDL)	7.34/35.26	0.02	99.8%	20.7%	$F_1 = 0.3356$ (0.7533 cell); burst $\rho = 0.438$	0.10
Reticulate Network Suite (400 sims, 45 splits)							
`N1` Strictly Clonal Null (100 + 200 AliSim)	Bushwhacker (MDL + Stage 4)	44.98/45.0	0.00	100.0%	99.96%	0.0% false lateral shunts (100% spec.)	0.09
`N3` Contiguous HGT Tract (450 bp)	Bushwhacker (MDL + Stage 4)	44.95/45.0	1.36	97.0%	96.9%	90.0% `INTERVAL` (IoU = 0.716; 0% false HGT clade)	0.12
`N4` Scattered Convergence (10 NT sites)	Bushwhacker (MDL + Stage 4)	44.95/45.0	0.00	100.0%	99.9%	100.0% `SPARSE`; 99.2% site prec.; 98.5% rec.	0.10
Single-Cell Somatic Suite (400 matrices)							
`C1` Binary SNV Clonal Burst (5% dropout)	Bushwhacker (`-discrete` MDL)	33.34/33.47	4.56	86.1%	85.6%	6.3× lower FP than binary ML (nRF = 0.1418)	0.52
	IQ-TREE 2 (Binary `GTR2/BIN`)	61.00/33.47	28.90	52.6%	95.8%	nRF = 0.4671	1.41
`C2` High Allelic Dropout (20% missing)	Bushwhacker (`-discrete` MDL)	36.41/37.15	13.33	63.2%	61.9%	nRF = 0.3725 (vs. 0.4829 IQ-TREE BIN)	0.57
`C3` Ordered Copy-Number ($K = 7$ states 0..6)	Bushwhacker (`-ordered` L_1 step)	59.60/55.60	5.61	90.6%	97.1%	nRF = 0.0660 (47% lower RF vs. binarized)	0.88
	Binarized Copy-Number Baseline	52.37/55.60	5.15	90.1%	84.9%	nRF = 0.1224 (RF = 13.53 vs. 7.22)	0.88
`C4` Noisy Diploid Null (100 controls)	Bushwhacker (`-ordered` MDL)	0.00/0.00	0.00	100.0%	100.0%	0.0 false clones across all 100 controls	0.55

Table S3: **Conditional local posterior branch probability calibration across 44,338 inferred branches without bootstrap resampling.** Conditional local posterior probability $P_{\text{local}}(e \mid T_{-e}, A) = (1 + e^{-G_{\text{MDL}}(e)})^{-1}$ is compared directly against ground-truth bipartition membership across multi-alphabet benchmarks (overall mean predicted $P_{\text{local}} = 98.08\%$, empirical true fraction = 94.35%, Expected Calibration Error ECE = 3.74%). Calibration is tight in the [0.99, 1.00] bin (89.4% of branches) and optimistic below 0.99 because P_{local} conditions on T_{-e}.

Conditional Local Posterior Bin $P_{\text{local}}(e \mid T_{-e}, A)$	Branches (n)	Mean Predicted P_{local}	Empirical True Fraction	Absolute Gap
[0.50, 0.65)	882	57.60%	40.59%	17.01%
[0.65, 0.80)	959	72.86%	48.80%	24.06%
[0.80, 0.90)	807	85.10%	57.25%	27.85%
[0.90, 0.95)	718	92.66%	69.36%	23.30%
[0.95, 0.99)	1,353	97.35%	80.27%	17.08%
[0.99, 1.00]	39,619	99.98%	98.33%	1.65%
Total / Weighted Summary	**44,338**	**98.08%**	**94.35%**	**ECE = 3.74%**

Table S4: **Ground-truth accuracy across 27,500 empirical-parameterized nucleotide histories from the Abadi et al. (*14*) and Gerth (*35*) archive.** Panel A stratifies `sim_bic` ($n = 5,901$) by short ancestral branches ($\ell_e L < 1.0$ expected substitution). Panel B reports homogeneous (`c0`), site-heterogeneous (`c1`), and branch-heterogeneous (`c2`) corpora.

Benchmark Stratum	Method	Res / True	RF_{bin}	nRF_{bin}	$RF_{\geq 1}$	$nRF_{\geq 1}$	FP / Tree	Precision
Panel A: `sim_bic` All ($n = 5,901$)	Bushwhacker (MDL)	8.9/18.1	10.51	0.2727	6.87	0.2303	0.68	94.7%
	IQ-TREE 2 (BIC ModelFinder)	18.1/18.1	8.42	0.1666	5.75	0.1616	4.21	83.3%
	IQ-TREE 2 (GTR + I + G4)	18.1/18.1	8.46	0.1678	5.77	0.1623	4.23	83.2%
	IQ-TREE 2 (Jukes–Cantor)	18.1/18.1	8.99	0.1782	6.33	0.1739	4.49	82.2%
	IQ-TREE 2 (`-polytomy`)	15.4/18.1	6.82	0.1404	3.34	0.1147	2.08	89.2%
	FastTree 2	16.7/18.1	7.24	0.1542	4.56	0.1516	2.93	84.9%
	BioNJ	16.7/18.1	11.57	0.2642	8.74	0.2710	5.10	73.1%
Low Stemminess (25–50% $\ell_e L < 1$, $n = 662$)	Bushwhacker (MDL)	10.7/22.0	11.83	0.2744	3.72	0.1522	0.29	98.1%
	IQ-TREE 2 (`-polytomy`)	14.2/22.0	12.05	0.2679	4.38	0.1650	2.14	86.9%
	FastTree 2	20.5/22.0	15.20	0.3306	10.23	0.3150	6.88	68.1%
	IQ-TREE 2 (BIC / GTR + I + G4)	22.0/22.0	16.51	0.3535	11.57	0.3388	8.26	64.7%
Very Low Stemminess (> 50% $\ell_e L < 1$, $n = 625$)	Bushwhacker (MDL)	7.8/33.6	26.14	0.3961	3.69	0.1797	0.18	98.8%
	FastTree 2	22.3/33.6	33.95	0.4992	15.12	0.4964	11.34	51.4%
	IQ-TREE 2 (`-polytomy`)	20.9/33.6	36.27	0.5388	14.20	0.4540	11.79	50.2%
	IQ-TREE 2 (BIC / GTR + I + G4)	33.6/33.6	45.23	0.6623	26.47	0.6352	22.62	33.8%
Panel B: `c0` Homogeneous ($n = 7,199$)	Bushwhacker (MDL)	8.7/16.8	13.30	0.3328	9.21	0.3074	2.60	82.5%
	IQ-TREE 2 (`-polytomy` / GTR + I + G4)	14.3/16.8	14.70/16.43	0.3015/0.3337	10.79/12.97	0.2865/0.3235	6.14/8.21	71.6%/66.6%
`c1` Site-Heterogeneous ($n = 7,200$)	Bushwhacker (MDL)	8.8/16.8	10.19	0.2578	6.11	0.2247	1.09	92.6%
	IQ-TREE 2 (`-polytomy` / GTR + I + G4)	14.2/16.8	10.30/12.04	0.1843/0.2158	6.48/8.89	0.1647/0.2133	3.85/6.02	84.9%/78.4%
`c2` Branch-Heterogeneous ($n = 7,200$)	Bushwhacker (MDL)	8.8/16.8	10.25	0.2616	6.16	0.2306	1.12	92.4%
	IQ-TREE 2 (`-polytomy` / GTR + I + G4)	14.2/16.8	10.36/12.08	0.1867/0.2180	6.55/8.92	0.1682/0.2163	3.91/6.04	84.6%/78.2%

Table S5: **Worked 4-taxon, 16-site MDL ledger and representative `AliSim` operating characteristics.** Panel A gives exact nat-by-nat code lengths on a quartet ($N = 4, L = 16$, $m_c = 2$ private leaf edits costing $\mathcal{L}_{\text{branch}}(2, 16) = 9.71$ nats each; matching Fig. S1). Panel B reports representative `AliSim` regimes (10 replicates each across 22 scenarios; 220 alignments total (*34*, *72*)).

Panel A: 4-Taxon Quartet Ledger ($L = 16$)	Shared Synapomorphies	Leaf Savings $\Delta\mathcal{L}_{\text{children}}$	Branch Code $\mathcal{L}_{\text{branch}}$	Topo Cost $\mathcal{L}_{\text{topo}}(2,2)$	Net Gain $G_{\text{MDL}}(e)$	Local Post. P_{local}	Decision
$m_e = 0$ (Zero shared mutations)	. . . | 2 priv.	0.00 nats	−0.69 nats	−1.10 nats	−1.79 nats	0.143	Contract (T_{star})
$m_e = 1$ (Single shared site)	G . . | 2 priv.	+6.55 nats	−5.56 nats	−1.10 nats	−0.11 nats	0.473	Contract (T_{star})
$m_e = 2$ (Two shared sites)	G G . | 2 priv.	+12.02 nats	−9.71 nats	−1.10 nats	+1.21 nats	0.771	Resolve (T_1, T_2)
$m_e = 3$ (Three shared sites)	G G G | 2 priv.	+16.69 nats	−12.99 nats	−1.10 nats	+2.60 nats	0.931	Resolve (T_1, T_2)
Panel B: AliSim Regime (N, L)	**Method**	**Res / True**	**FP**	**Precision**	**RF**	**nRF / FPR**	**Time (s)**
`S08` Star Null (128 × 1,000 bp)	Bushwhacker (MDL)	0.0/0	0.0	100.0%	0.0	0.0% FPR	0.55
	IQ-TREE 2 (`-polytomy` / Binary)	74.8/125.0	74.8/125.0	0.0%	74.8/125.0	59.8%/100% FPR	482.5/391.6
	FastTree 2 (raw / SH≥ 0.90)	124.6/11.9	124.6/11.9	0.0%	124.6/11.9	99.7%/9.5% FPR	3.02
`S08b` Ultra-Short (200 × 100 bp)	Bushwhacker (MDL)	16.3/197	0.1	99.7%	180.9	0.4591	0.37
	IQ-TREE 2 (`-polytomy` / Binary)	142.4/197.0	31.3/74.0	78.1%/62.4%	117.2/148.0	0.2975/0.3756	96.2/91.1
`S10b` Standard (200 × 1,000 bp)	Bushwhacker (MDL / Binary)	157.2/197.0	0.0/8.6	100.0%/95.6%	39.8/17.2	0.1010/0.0437	0.63
	IQ-TREE 2 (Binary)	197.0/197	9.0	95.4%	18.0	0.0457	65.42
`S12c` Long Gene (200 × 16,000 bp)	Bushwhacker (MDL)	196.0/197	0.0	100.0%	1.0	0.0025	2.65
`S04` Large Tree (1,024 × 1,000 bp)	Bushwhacker (MDL)	784.9/1,021	0.1	99.99%	236.3	0.1157	4.22
	IQ-TREE 2 (Binary) / BioNJ	1,021.0/1,021	50.1/134.2	95.1%/86.9%	100.2/268.4	0.0491/0.1314	17.85/0.10
`S13` Low-Div Outbreak (200 × 1,000 bp)	Bushwhacker (MDL)	102.5/197	0.0	100.0%	94.5	0.2398	0.32
	IQ-TREE 2 (Binary) / FastTree 2	197.0/196.9	33.8/32.9	82.8%/83.3%	67.6/65.9	0.1716/0.1673	62.5/3.36

Table S6: **Multi-seed maximum-likelihood optimization instability and apparent species-tree conflict removal across eukaryotic phylogenomic radiations.** Top rows report search-unstable ML splits across 20 independent RAxML/IQ-TREE optimization starts per gene on 4,159 gene-alignments (3,315 unique genes; `WickD3a`/`WickD3b` paired codon views) (83,180 searches (*24*)). Bottom summary encompasses all 19 benchmarks (45,205 loci (*13*, *24*)).

Dataset (n_{genes}, $\bar{N}$, $\bar{L}$)	Mean $\rho(A)$	Search-Unstable ML Splits	Contracted by BW (MDL)	Method	Res / Sp	Discordant Splits	Concordance Prec. (nRF_{sp})
`SongD1` Mammals (424, 37, 3,157 bp)	12.53%	1,715/14,416 (11.9%)	1,612/1,715 (94.0%)	Bushwhacker / Best-40 ML	22.3/34.0	6.82/9.04	69.4%/73.4%
`XiD4` Seed Plants (310, 33, 773 bp)	8.62%	2,715/9,357 (29.0%)	2,555/2,715 (94.1%)	Bushwhacker / Best-40 ML	16.9/30.2	7.46/13.63	56.7%/55.3%
`PrumD6` Birds (259, 199, 1,524 bp)	36.92%	18,815/50,762 (37.1%)	16,188/18,815 (86.0%)	Bushwhacker / Best-40 ML	141.4/196.0	41.53/73.54	70.7%/62.4% (0.352 vs. 0.376)
`WickD3a` Plants 1+2+3 (844, 49, 2,000 bp)	9.60%	8,378/38,638 (21.7%)	8,101/8,378 (96.7%)	Bushwhacker / Best-40 ML	19.9/45.8	11.26/19.58	46.9%/57.9%
`WickD3b` Plants 1+2 (844, 49, 1,333 bp)	14.11%	11,118/38,638 (28.8%)	10,504/11,118 (94.5%)	Bushwhacker / Best-40 ML	24.4/45.8	13.81/21.55	45.5%/54.1%
`MisoD2a` Insects (1,478, 132, 1,208 bp)	8.87%	122,160/190,030 (64.3%)	116,861/122,160 (95.7%)	Bushwhacker / Best-40 ML	75.5/128.6	58.79/81.40	23.0%/36.4%
All 6 Multi-Search Sets (4,159 alns, 3,315 genes)	**12.18%**	**164,901/341,841 (48.2%)**	**155,821/164,901 (94.5%)**	Bushwhacker / Best-40 ML	48.2/82.2	29.82/43.79	42.6%/51.2%
All 19 Eukaryotic Sets (45,205 loci)	—	9 AA + 10 NT clades	61.5% sp. conflicts cut	1,522,139 → 585,973	936,166 cut	74.9% birds; 82.5% animals	20.1–20.2× AA speedup

Table S7: **Proteome-wide mammalian concordance, cochlear and ecological convergence (**$19,952$ **TOGA2 loci across** 882 **mammals), and pandemic viral recombinant mapping.** Panel A summarizes mammalian coding-DNA and 20-state protein concordance (*28*, *47*). Panel B highlights certified mammalian adaptive convergence, tandem paralog conversion, and unsupervised viral recombinants. In Panel A, `Cand. / Ret. Conflict%` gives pooled discordant splits divided by candidate binary splits (14.94M) and by retained splits, respectively (with per-gene mean retained conflict in parentheses where computed).

Panel A: Mammalian Proteome Mode	Alignment / Tree Mode	Loci (n)	Total Splits	Sp. Shared	Sp. Conflict	Cand. / Ret. Conflict%	Conflicts Removed	Certified Shunts (`INT` / `SPARSE`)
Coding DNA	Forced-binary tree	19,890	14,936,938	7,960,411	6,976,527	46.7%/46.7% (48.5%)	Baseline	—
(med. $N = 855$, $L = 1{,}383$ bp)	MDL-contracted tree	19,890	9,876,602	6,774,016	3,102,586	20.8%/31.4% (33.8%)	−3,873,941 (55.53%)	1,032 / 14,105 (free tree)
20-State Protein (`-translate`)	MDL-contracted (`AA`)	19,952	5,226,678	2,274,433	2,952,245	19.8%/56.5%	−7,871,929 (72.73%)	4,202 / 11,713 (species tree)
Matched Alignment Subset	De Novo Codon MSA (`VALRUS`)	5,058	1,537,531	1,135,904	401,627	21.7%/26.1% (28.7%)	−107,754 (21.15%)	200 / 1,671 (−25.3% splice noise)

Panel B: Certified Target / Cohort	Biological System / Guild	$N \times L$	Mode	Shunts	Top Gain (nats)	Statistical Enrichment / Runtime	Primary Certified Tract or Convergent Signal
Cochlear Complex (16/25 genes)	Mammalian auditory hair cell	819–875 sp.	AA + NT	48 `SPARSE`	+54.61 max AA (`PCDH15`; +93.2 cum.)	64% of core complex	`TMC1`, `CDH23`, `LOXHD1`, `SLC26A5` (+6.64/ + 2.79 NT; +4.0 AA)
Myrmecophagous Guild	Pangolins ↔ anteaters/armadillos	882 sp. AA	20-state AA	18 (13 fam.)	+113.5 (`OR10D3`)	18.6× fam. ($p = 8.1 \times 10^{-13}$, $q = 3.7 \times 10^{-12}$); 25.8× raw	`OR10D3`, `TTN` (+42.4), `ALMS1` (+28.6)
Subterranean Blind Guild	Golden/true moles, mole-rats	882 sp. AA	20-state AA	6 (6 fam.)	+27.1 (`EPAS1`)	4.0× O/E ($p = 0.0045$, $q = 0.0057$)	Subterranean hypoxia (`EPAS1`, `EGLN1`) & sensory remodeling
Bat–Whale Sonar Screen	Echolocating bats ↔ whales	882 sp. AA	20-state AA	53 cross-order	+66.2 (`ORL1G2`)	0.43× O/E (123.8 exp.)	Proteome-wide neutral; sonar concentrated in cochlear complex
Tandem Paralog Arrays	Keratin, `TRIM`, ZNF, Globins	92–880 sp.	`INTERVAL`	4,202 AA	+418.9 (`KRT31`)	94.2% intra-order (3.4–4.5×)	Concerted conversion (`TRIM22` +313.5, `CEACAM1/7`, `HBA1/2`)
SARS-CoV-2 Global + Blinded `X*`	Pandemic coronavirus	10,062 × 29,903 bp	NT + Stage 4	3 + 6 `INT`	+52.61 / +31.41	165.4 s (> 28.7× vs. 4,758 s FT timeout); 1.18 s `X*`	Spike RBD [21,329, 22,917], ORF6–7a, `XBF/XBT/XAG/XBB`
Global Dengue (`DENV-1..4`)	Urban arbovirus serotypes	5,498 genomes	NT + AA	14 `INTERVAL`	+77.08 (`DENV-2`)	10.2–28.5 s (12.9–22.6× FT)	NS2A [3,632, 3,717], Capsid [8, 238] mosaic tracts
`HIV-1` `KAL153` (`CRF03_AB`)	Circulating recombinant HIV	9 × 9,953 bp	NT + Stage 4	2 `INTERVAL`	+29.23	0.141 s (75,950× `GARD`)	Exact `gag` [5, 2,598] and `nef` [9,102, 9,757] tracts

Table S8: **Domain-wide prokaryotic marker analysis and quantitative test of the Complexity Hypothesis across** 242 **GTDB Release 95 markers (*29*, *30*, *55*).** Panel A reports single-gene nucleotide topological compatibility across $1,672$ Archaea and $1,000$ Bacteria (111 phyla). Panel B stratifies functional classes in nucleotide and native 20-state amino-acid space.

Panel A: Domain Stratum	Method	Res / Sp	Discordant FP	Prec.	RF (nRF)	Time (s)	Phylum Rec / Comp	Order Rec / Comp	Genus Rec / Comp
`bac120` Bacteria (120 genes, 111 phyla)	Bushwhacker (MDL)	335.6/857.8	192.6	42.7%	907.1 (0.5290)	23.03	13.8%/27.7%	28.0%/63.3%	67.6%/92.3%
	FastTree 2 (GTR + Γ)	857.6/857.8	495.5	42.2%	991.0 (0.5777)	20.36	29.6%/29.7%	59.0%/59.0%	86.0%/86.0%
`ar122` Archaea (122 genes, 19 phyla)	Bushwhacker (MDL)	855.2/1,387.4	449.6	51.1%	1,432.7 (0.4927)	16.04	22.4%/28.5%	38.0%/52.6%	68.1%/82.0%
	FastTree 2 (GTR + Γ)	1,387.2/1,387.4	624.1	55.0%	1,418.2 (0.4502)	25.20	33.4%/33.4%	55.0%/55.0%	81.6%/81.6%

Panel B: Complexity Hypothesis Class	Markers (n)	Mean $\bar{L}$	Mean $p(A)$	BW Prec.	Inter-Phylum HGT/Gene	Total NT Jumps/Gene	`ar122` 20-State AA Shunt Loci	Cumulative AA Shunt Gain	Fisher Test
Informational Core (Ribosome / RNAP)	85 NT / 20 AA	613 bp	31.2%	48.2%	13.81	25.08	0/20 (0.0%)	0.0 nats	$p = 0.0235$, $q = 0.027$
Transcription & DNA Replication	43 NT	1,307 bp	25.6%	46.7%	14.14	24.74	—	—	$p = 3.2 \times 10^{-4}$
Operational Metabolic & Cellular	83 NT / 102 AA	938 bp	26.8%	45.9%	14.58	24.11	21/102 (20.6%, 30 shunts)	391.8 nats	(3.7× NT op. HGT)
Aminoacyl-tRNA Synthetases	31 NT	1,649 bp	26.0%	43.3%	15.29	27.13	Included in operational AA (`pheS` +32.3)	DPANN Nanoarchaeota highways	

Table S9: **Lossless phylogenetic genome archiving (`.bwz`) compared against general-purpose dictionary compressors (`gzip -9`, `bzip2 -9`, `xz -6`/`lzma -9`) and mosaic recombination switchpoint titration.** Sizes are in bytes (verified bit-exact upon `bushwhacker -d` decompression); both sequence-payload bytes (`Payload`) and full self-contained archive bytes including taxa/Newick/headers (`Archive`) are reported, with Ratio $> 1.00\times$ indicating `.bwz` payload outperforms the best general-purpose compressor and Ratio $< 1.00\times$ flagging genome-wide mosaicism.

Dataset / Regime	Category	$N \times L$	Raw FASTA (B)	`gzip -9`	`bzip2 -9`	`xz -6 / lzma -9`	`.bwz` Payload (B)	`.bwz` Archive (B)	Payload Ratio vs. Best GP (`gzip` ratio)
`SARS-CoV-2_10062`	Pandemic Viral	10,062 × 29,903	300,883,986	7,322,102	2,641,139	797,044	387,697 (38.5 B/g)	635,851 (63.2 B/g)	2.06× (18.89× gz; arch. 1.25×)
`SARS-CoV-2_5097`	Pandemic Viral	5,097 × 29,903	152,415,591	3,791,027	1,312,281	409,108	220,588 (43.3 B/g)	340,570 (66.8 B/g)	1.86× (17.19× gz; arch. 1.20×)
`DENV-1..4` (4 cohorts)	Pandemic Arboviral	5,498 × 10,188	56,013,624	5.2–6.4× `.bwz`	—	1.26–1.52× `.bwz`	—	5.23–6.42× < gzip	1.26–1.52×
`Influenzaa.fasta`	Seasonal Viral	349 × 987	344,463	6,729	4,813	5,005	4,133	10,371	1.17× (1.63× gzip)
`Human_mtDNA`	Organelle Genome	78 × 16,585	1,293,630	23,424	9,709	9,348	6,051	8,353	1.55× (3.87× gzip)
`prestin.fasta`	Mammalian Gene	116 × 2,235	259,260	27,291	19,154	11,821	7,502	10,578	1.58× (3.64× gzip)
`rbcL_483`	Deep Plastid	483 × 1,398	675,234	54,956	44,819	31,835	22,052	33,460	1.44× (2.49× gzip)
`16S_N2048` / `N32768`	Microbiome 16S	2,048–32,768 × 1,348	2,760,704+	406,975+	277,753	148,878	106,817	156,355 (64.4 B@32k)	1.39× (1.08× xz @32k)
`PMEN1` (24.4 Mbp)	Bacterial Chrom.	11 × 2,221,315	24,434,465	6,434,271	706,678	599,856	594,506	594,774	1.01× (10.82× gzip)
`Hominoidea_MHC` / `pBP2x`	Local Conv. / Mosaic	379–12	427,512/248,196	9,944/11,707	7,438/8,131	6,402/7,144	4,176/5,928	12,720/6,169	1.53× / 1.21×
`Sarbeco_83` (31 kb)	Whole-Genome Mosaic	83 × 30,965	2,570,095	222,223	103,100	78,441	110,499	114,717	0.71× (Mosaic switch)
`HIV1_CRF02_AG`	Whole-Genome Mosaic	9 × 9,953	89,577	16,121	10,798	9,524	14,768	15,018	0.65× (Mosaic switch)
Titration $K = 0 \to 1 \to 8 \to 32$	0–32 bkpts/mosaic	56 × 12,000	672,000	168.4–180.4k	94.4–102.6k	54.4 → 59.5 → 55.8 → 57.4k	43.7 → 47.3 → 50.0 → 59.6k	45.2 → 48.7 → 51.9 → 61.9k	1.24× → 1.26× → 1.12× → 0.96×

Table S10: **Zero-leakage single-cell human tumor phylogenomics across** $1,001,609$ **myeloid cells (ordered 3-state somatic zygosity) and** $4,950$ **solid-tumor single-cell whole-genome copy-number profiles (ordered 12-state CNA).** All cell counts, anatomic labels, and passage numbers were withheld during tree and network inference.

Cohort / Modality	Representation (`-ordered`)	Cells / Runs	Features (L)	Resolved / Max Splits	Collapsed%	Compression	Runtime (s)	Post-Hoc Biological Validation
COMPASS Myeloid scDNA (*61*)	Unthresholded 3-state zygosity	1,001,609 (168 runs)	3–28 driver loci	6,222 unique clones	Burst-calibrated	—	2.62 s	$k \geq 3$ hubs 6.18× cells (run $p = 1.7 \times 10^{-18}$); 90.3% CN-LOH
Morita AML Tapestri (*32*)	Curated + longitudinal zygosity	123 patients (465 clones)	31 driver genes	100 branching pts	Multifurcating	`AML-38-001` 3.0 ms	0.128 s	Signaling OR = 1.79 subclonal ($p = 1.9 \times 10^{-5}$); 44 pts parallel
`OV2295` HGSOC (*62*)	Colinear segments / Raw 500-kb bins	1,111 cells	420 segs / 4,375 bins	631/1,108 (648 bin)	43.1% (41.5%)	6.0× / 29.4×	4.66 s / 20.75 s	Ovary vs. ascites $F_1 = 0.997/0.995$ (95.7% pur.); $\rho = 0.818$
`SA501` TNBC PDX (*62*)	Colinear segments / Raw 500-kb bins	2,691 cells	365 segs / 4,375 bins	1,639/2,688 (1,799 bin)	39.0% (33.1%)	6.5× / 33.5×	11.30 s / 65.53 s	Multi-passage sweeps (`X2` → `X15`); 60.7% bin purity
`SA609` TNBC PDX (*62*)	Colinear segments / Raw 500-kb bins	600 cells	448 segs / 4,375 bins	280/597 (299 bin)	53.1% (49.9%)	4.2× / 21.9×	4.68 s / 19.97 s	Diploid normal (`SA928`, 91.3% dominant clade) vs. `X2`
`SA535` TNBC PDX (*62*)	Colinear segments / Raw 500-kb bins	548 cells	426 segs / 4,375 bins	263/545 (285 bin)	51.7% (47.7%)	6.0× / 29.8×	1.26 s / 7.35 s	Subclonal passage clades (91.5% pur.; 21.9 s seg; 113.6 s bin)

Table S11: **Continuous-time reversible hard-CAT rate-category compression (`GTR+CAT8` and `LG08+CAT8`) with 64-site cache-tiled pruning across Mammalian, Archaeal, and Bacterial Tree-of-Life supermatrices (all methods benchmarked on 32 CPU cores).**

Tree-of-Life Supermatrix	Method (32 Threads)	Runtime (s)	Speedup	Peak RAM (GB)	Prec. / Rec. (nRF)	Rank 1 (Superord./Phylum)	Rank 2 (Order/Class)	Rank 3 (Family/Order)	Rank 4 (Genus/Family)
Mammalia Coding DNA	**Bushwhacker (`GTR+CAT8` MDL)**	**48.59**	**1.0**×	**2.39**	**91.9**%/**91.8**% (**0.081**)	**100.0**% (4/4)	**100.0**% (18/18)	**95.5**% (84/88)	**94.0**% (172/183)
(882 sp. ×543,864 bp,	IQ-TREE 2 (`GTR+G4 -fast`)	1,166.56	24.0× slower	39.82 (16.7×)	92.4%/92.4% (0.076)	100.0%	100.0%	95.5%	94.5%
253 single-copy loci; 92.7% BW–IQ)	FastTreeMP (`GTR+CAT`)	13,990.59	287.9× slower	—	92.6%/92.6% (0.074)	100.0%	100.0%	95.5%	94.5%
Mammalia Protein	**Bushwhacker (`LG08+CAT8` MDL)**	**65.78**	**1.0**×	**1.98**	**95.8**%/**83.6**% (**0.100**; 32 FP)	**100.0**% (4/4)	**100.0**% (18/18)	**95.5**% (84/88)	**92.4**% (169/183)
(882 sp. ×181,288 AA,	IQ-TREE 2 (`LG+G4 -fast`)	2,434.49	37.0× slower	79.15 (40.0×)	91.6%/91.6% (0.084; 74 FP)	100.0%	100.0%	95.5%	93.4%
112 silent AA stems contracted)	FastTreeMP (`LG+CAT`)	12,711.41	193.2× slower	—	91.4%/91.4% (0.086; 76 FP)	100.0%	100.0%	95.5%	93.4%
Archaea Domain (`ar122`)	**Bushwhacker (`LG08+CAT8` MDL)**	**31.50**	**1.0**×	**1.40**	**84.4**%/**84.4**% (**0.156**)	**82.4**% (14/17)	**90.0**% (36/40)	**98.8**% (84/85)	**98.6**% (146/148)
(1,672 sp. ×32,675 AA,	IQ-TREE 2 (`LG+G4 -fast`)	1,691.53	53.7× slower	32.95 (23.5×)	86.9%/86.9% (0.131)	88.2%	95.0%	98.8%	100.0%
19 phyla, ∼ 3.5 Gyr)	FastTreeMP (`LG+CAT`)	4,499.48	142.8× slower	—	86.5%/86.5% (0.135)	88.2%	92.5%	97.7%	99.3%
	Bushwhacker v0 (Unweighted-Fitch Legacy)	202.32	6.4× slower	—	37.5%/37.2% (0.624)	5.9% (1/17)	7.5%	20.0%	28.4%
Bacteria Domain (`bac120`)	**Bushwhacker (`LG08+CAT8` MDL)**	**26.91**	**1.0**×	**0.95**	**75.3**%/**75.3**% (**0.247**)	**94.3**% (99/105)	**97.1**% (166/171)	**99.6**% (222/223)	**99.0**% (203/205)
(1,000 sp. ×43,733 AA,	IQ-TREE 2 (`LG+G4 -fast`)	1,137.85	42.3× slower	26.37 (27.8×)	76.4%/76.4% (0.236)	94.3%	98.3%	99.6%	98.5%
all 111 phyla, ∼ 3.5 Gyr)	FastTreeMP (`LG+CAT`)	3,822.63	142.1× slower	—	77.2%/77.2% (0.228)	89.5%	95.3%	99.6%	99.5%
	Bushwhacker v0 (Unweighted-Fitch Legacy)	291.21	10.8× slower	—	26.6%/26.4% (0.733)	15.2% (16/105)	30.4%	39.9%	43.9%

References and Notes

1. Walter M. Fitch. Toward defining the course of evolution: minimum change for a specific tree topology. *Systematic Zoology*, 20(4):406–416, 1971. doi: 10.2307/2412116.

2. Joseph Felsenstein. Evolutionary trees from DNA sequences: a maximum likelihood approach. *Journal of Molecular Evolution*, 17(6):368–376, 1981. doi: 10.1007/BF01734359.

3. Naruya Saitou and Masatoshi Nei. The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution*, 4(4):406–425, 1987. doi: 10.1093/oxfordjournals.molbev.a040454.

4. Morgan N. Price, Paramvir S. Dehal, and Adam P. Arkin. FastTree 2 – approximately maximum-likelihood trees for large alignments. *PLoS ONE*, 5(3):e9490, 2010. doi: 10.1371/journal.pone.0009490.

5. Alexey M. Kozlov, Diego Darriba, Tomáš Flouri, Benoit Morel, and Alexandros Stamatakis. RAxML-NG: a fast, scalable and user-friendly tool for maximum likelihood phylogenetic inference. *Bioinformatics*, 35(21):4453–4455, 2019. doi: 10.1093/bioinformatics/btz305.

6. Bui Quang Minh, Heiko A. Schmidt, Olga Chernomor, Dominik Schrempf, Michael D. Woodhams, Arndt von Haeseler, and Robert Lanfear. IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Molecular Biology and Evolution*, 37(5): 1530–1534, 2020. doi: 10.1093/molbev/msaa015.

7. Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14(5):465–471, 1978. doi: 10.1016/0005-1098(78)90005-5.

8. Jorma Rissanen. A universal prior for integers and estimation by minimum description length. *The Annals of Statistics*, 11(2):416–431, 1983. doi: 10.1214/aos/1176346150.

9. Joseph Felsenstein. Confidence limits on phylogenies: an approach using the bootstrap. *Evolution*, 39(4):783–791, 1985. doi: 10.1111/j.1558-5646.1985.tb00420.x.

10. David Posada and Thomas R. Buckley. Model selection and model averaging in phylogenetics: advantages of Akaike information criterion and Bayesian approaches over likelihood ratio tests. *Systematic Biology*, 53(5):793–808, 2004. doi: 10.1080/10635150490522304.

11. Sergei L. Kosakovsky Pond, David Posada, Michael B. Gravenor, Christopher H. Woelk, and Simon D. W. Frost. Automated phylogenetic detection of recombination using a genetic algorithm. *Molecular Biology and Evolution*, 23(10):1891–1901, 2006. doi: 10.1093/molbev/msl051.

12. Antonis Rokas and Sean B. Carroll. Bushes in the tree of life. *PLoS Biology*, 4(11):e352, 2006. doi: 10.1371/journal.pbio.0040352.

13. Xing-Xing Shen, Chris Todd Hittinger, and Antonis Rokas. Contentious relationships in phylogenomic studies can be driven by a handful of genes. *Nature Ecology & Evolution*, 1(5): 0126, 2017. doi: 10.1038/s41559-017-0126.

14. Shiran Abadi, Dana Azouri, Tal Pupko, and Itay Mayrose. Model selection may not be a mandatory step for phylogeny reconstruction. *Nature Communications*, 10(1):934, 2019. doi: 10.1038/s41467-019-08822-w.

15. Joseph Felsenstein. Cases in which parsimony or compatibility methods will be positively misleading. *Systematic Zoology*, 27(4):401–410, 1978. doi: 10.2307/2412923.

16. Simon Tavaré. Some probabilistic and statistical problems in the analysis of DNA sequences. In *Lectures on Mathematics in the Life Sciences*, volume 17, pages 57–86. American Mathematical Society, Providence, RI, 1986.

17. Ziheng Yang. Maximum likelihood phylogenetic estimation from DNA sequences with variable rates over sites: approximate methods. *Journal of Molecular Evolution*, 39(3):306–314, 1994. doi: 10.1007/BF00160154.

18. Si Quang Le and Olivier Gascuel. An improved general amino acid replacement matrix. *Molecular Biology and Evolution*, 25(7):1307–1320, 2008. doi: 10.1093/molbev/msn067.

19. W. Ford Doolittle. Phylogenetic classification and the universal tree. *Science*, 284(5423): 2124–2128, 1999. doi: 10.1126/science.284.5423.2124.

20. Michael Worobey, Andrew Rambaut, and Edward C. Holmes. Widespread intra-serotype recombination in natural populations of dengue virus. *Proceedings of the National Academy of Sciences*, 96(13):7352–7357, 1999. doi: 10.1073/pnas.96.13.7352.

21. Peter Parham and Tomoko Ohta. Population biology of antigen presentation by MHC class I molecules. *Science*, 272(5258):67–74, 1996. doi: 10.1126/science.272.5258.67.

22. Yang Liu, James A. Cotton, Bin Shen, Xiuqun Han, Stephen J. Rossiter, and Shuyi Zhang. Convergent sequence evolution between echolocating bats and dolphins. *Current Biology*, 20 (2):R53–R54, 2010. doi: 10.1016/j.cub.2009.11.058.

23. Ying Li, Zhen Liu, Peng Shi, and Jianzhi Zhang. The hearing gene Prestin unites echolocating bats and whales. *Current Biology*, 20(2):R55–R56, 2010. doi: 10.1016/j.cub.2009.11.042.

24. Xiaofan Zhou, Xing-Xing Shen, Chris Todd Hittinger, and Antonis Rokas. Evaluating fast maximum likelihood-based phylogenetic programs using empirical phylogenomic data sets. *Molecular Biology and Evolution*, 35(2):486–503, 2018. doi: 10.1093/molbev/msx302.

25. Joe Parker, Georgia Tsagkogeorga, James A. Cotton, Yuan Liu, Paolo Provero, Elia Stupka, and Stephen J. Rossiter. Genome-wide signatures of convergent evolution in echolocating mammals. *Nature*, 502(7470):228–231, 2013. doi: 10.1038/nature12511.

26. Zhengting Zou and Jianzhi Zhang. No genome-wide protein sequence convergence for echolocation. *Molecular Biology and Evolution*, 32(5):1237–1241, 2015. doi: 10.1093/molbev/msv014.

27. Yatish Turakhia, Bryan Thornlow, Angie Hinrichs, Jakob McBroome, Nicolas Ayala, Cheng Ye, Kyle Smith, Nicola De Maio, David Haussler, Robert Lanfear, and Russell Corbett-Detig. Pandemic-scale phylogenomics reveals the SARS-CoV-2 recombination landscape. *Nature*, 609(7929):994–997, 2022. doi: 10.1038/s41586-022-05189-9.

28. Yury V. Malovichko, Bernhard Bein, Alejandro Gonzales-Irribarren, Evgeny Leushkin, Leon Hilgers, Amy Stephen, Xueling Yi, Michele Albertini, Tim Stadager, Markus Zumpt, Luca Hoppach, Felix Götz, Niklas Himstedt, Lucas Koch, Vertebrate Genomes Project Consortium Phase I, and Michael Hiller. Accurate, comprehensive gene annotation and ortholog identification across thousands of vertebrate genomes with TOGA2. *bioRxiv*, 2026. doi: 10.64898/2026.06.30.735536.

29. Ravi Jain, Maria C. Rivera, and James A. Lake. Horizontal gene transfer among genomes: the complexity hypothesis. *Proceedings of the National Academy of Sciences*, 96(7):3801–3806, 1999. doi: 10.1073/pnas.96.7.3801.

30. Donovan H. Parks, Maria Chuvochina, Pierre-Alain Chaumeil, Christian Rinke, Aaron J. Mussig, and Philip Hugenholtz. A complete domain-to-species taxonomy for Bacteria and Archaea. *Nature Biotechnology*, 38(9):1079–1086, 2020. doi: 10.1038/s41587-020-0501-8.

31. William S. DeWitt, Ashni A. Vora, Tatsuya Araki, Jared G. Galloway, Tanwee Alkutkar, Juliana Bortolatto, Tiago B. R. Castro, Will Dumm, Chris Jennings-Shaffer, Tongqiu Jia, Luka Mesin, Jesse D. Bloom, Gabriel Ozorowski, Juhee Pae, Duncan K. Ralph, Armita Nourmohammad, Yun S. Song, Andrew B. Ward, Tyler N. Starr, Frederick A. Matsen, IV, and Gabriel D. Victora. Replaying germinal center evolution on a quantified affinity landscape. *Cell*, 2026.

32. Kiyomi Morita, Feng Wang, Katharina Jahn, Tianyuan Hu, Tomoyuki Tanaka, Yuya Sasaki, Jack Kuipers, Sanam Loghavi, Sa A. Wang, Yan Yan, Ken Furudate, Jairo Matthews, Latasha Little, Curtis Gumbs, Jianhua Zhang, Xingzhi Song, Erika Thompson, Keyur P. Patel, Carlos E. Bueso-Ramos, Courtney D. DiNardo, Farhad Ravandi, Elias Jabbour, Michael Andreeff, Jorge Cortes, Kapil Bhalla, Guillermo Garcia-Manero, Hagop Kantarjian, Marina Konopleva, Daisuke Nakada, Nicholas Navin, Niko Beerenwinkel, P. Andrew Futreal, and Koichi Takahashi. Clonal evolution of acute myeloid leukemia revealed by high-throughput single-cell genomics. *Nature Communications*, 11(1):5327, 2020. doi: 10.1038/s41467-020-19119-8.

33. Margaret O. Dayhoff, Robert M. Schwartz, and Bruce C. Orcutt. A model of evolutionary change in proteins. In Margaret O. Dayhoff, editor, *Atlas of Protein Sequence and Structure*, volume 5 of *Supplement 3*, pages 345–352. National Biomedical Research Foundation, Washington, DC, 1978.

34. Nhan Ly-Trong, Suha Naser-Khdour, Robert Lanfear, and Bui Quang Minh. AliSim: a fast and versatile phylogenetic sequence simulator for the genomic era. *Molecular Biology and Evolution*, 39(5):msac092, 2022. doi: 10.1093/molbev/msac092.

35. Michael Gerth. Neglecting model selection alters phylogenetic inference. *bioRxiv*, page 849018, 2019. doi: 10.1101/849018.

36. James H. Degnan and Noah A. Rosenberg. Discordance of species trees with their most likely gene trees. *PLoS Genetics*, 2(5):e68, 2006. doi: 10.1371/journal.pgen.0020068.

37. Chao Zhang, Rasmus Nielsen, and Siavash Mirarab. ASTER: A package for large-scale phylogenomic reconstructions. *Molecular Biology and Evolution*, 42(8):msaf172, 2025. doi: 10.1093/molbev/msaf172.

38. Erich D. Jarvis, Siavash Mirarab, Andre J. Aberer, Bo Li, Peter Houde, Cai Li, Simon Y. W. Ho, Brant C. Faircloth, Benoit Nabholz, Jason T. Howard, et al. Whole-genome analyses resolve early branches in the tree of life of modern birds. *Science*, 346(6215):1320–1331, 2014. doi: 10.1126/science.1253451.

39. Richard O. Prum, Jacob S. Berv, Alex Dornburg, Daniel J. Field, Jeffrey P. Townsend, Emily Moriarty Lemmon, and Alan R. Lemmon. A comprehensive phylogeny of birds (Aves) using targeted next-generation DNA sequencing. *Nature*, 526(7574):569–573, 2015. doi: 10.1038/nature15697.

40. Norman J. Wickett, Siavash Mirarab, Nam Nguyen, Tandy Warnow, Eric Carpenter, Naim Matasci, Saravanaraj Ayyampalayam, Michael S. Barker, J. Gordon Burleigh, Matthew A. Gitzendanner, et al. Phylotranscriptomic analysis of the origin and early diversification of land plants. *Proceedings of the National Academy of Sciences*, 111(45):E4859–E4868, 2014. doi: 10.1073/pnas.1323926111.

41. Bernhard Misof, Shanlin Liu, Karen Meusemann, Ralph S. Peters, Alexander Donath, Christoph Mayer, Paul B. Frandsen, Jessica Ware, Tomáš Flouri, Rolf G. Beutel, et al. Phylogenomics resolves the timing and pattern of insect evolution. *Science*, 346(6210):763–767, 2014. doi: 10.1126/science.1257570.

42. Meng-Yun Chen, Dan Liang, and Peng Zhang. Selecting question-specific genes to reduce incongruence in phylogenomics: a case study of jawed vertebrate backbone phylogeny. *Systematic Biology*, 64(6):1104–1120, 2015. doi: 10.1093/sysbio/syv059.

43. Marek L. Borowiec, Ernest K. Lee, Joanna C. Chiu, and David C. Plachetzki. Extracting phylogenetic signal and accounting for bias in whole-genome data sets supports the Ctenophora as sister to remaining Metazoa. *BMC Genomics*, 16:987, 2015. doi: 10.1186/s12864-015-2146-4.

44. Nathan V. Whelan, Kevin M. Kocot, Tatiana P. Moroz, Krishanu Mukherjee, Peter Williams, Gustav Paulay, Leonid L. Moroz, and Kenneth M. Halanych. Ctenophore relationships and their placement as the sister group to all other animals. *Nature Ecology & Evolution*, 1(11): 1737–1746, 2017. doi: 10.1038/s41559-017-0331-3.

45. Sen Song, Liang Liu, Scott V. Edwards, and Shaoyuan Wu. Resolving conflict in eutherian mammal phylogeny using phylogenomics and the multispecies coalescent model. *Proceedings of the National Academy of Sciences*, 109(37):14942–14947, 2012. doi: 10.1073/pnas.1211733109.

46. Allison J. Shultz and Timothy B. Sackton. Immune genes are hotspots of shared positive selection across birds and mammals. *eLife*, 8:e41815, 2019. doi: 10.7554/eLife.41815.

47. Bogdan M. Kirilenko, Chetan Munegowda, Ekaterina Osipova, David Jebb, Virag Sharma, Moritz Blumer, Ariadna E. Morales, AW Ahmed, Dimitrios-Georgios Kontopoulos, Leon Hilgers, Kerstin Lindblad-Toh, Elinor K. Karlsson, Zoonomia Consortium, and Michael Hiller. Integrating gene annotation with orthology inference at scale. *Science*, 380(6643): eabn3107, 2023. doi: 10.1126/science.abn3107.

48. Gregg W. C. Thomas and Matthew W. Hahn. Determining the null model for detecting adaptive convergence from genomic data: a case study using echolocating mammals. *Molecular Biology and Evolution*, 32(5):1232–1236, 2015. doi: 10.1093/molbev/msv013.

49. Pascal-Antoine Christin, Erika J. Edwards, Guillaume Besnard, Susanna F. Boxall, Richard Gregory, Elizabeth A. Kellogg, James Hartwell, and Colin P. Osborne. Adaptive evolution of C_4 photosynthesis through recurrent lateral gene transfer. *Current Biology*, 22(5):445–449, 2012. doi: 10.1016/j.cub.2012.01.054.

50. Jean K. Carr, Mika O. Salminen, Jan Albert, Eric Sanders-Buell, D. Gotte, Deborah L. Birx, and Francine E. McCutchan. Full genome sequences of human immunodeficiency virus type 1 subtypes G and A/G intersubtype recombinants. *Virology*, 247(1):22–31, 1998. doi: 10.1006/viro.1998.9211.

51. Ian H. Witten, Radford M. Neal, and John G. Cleary. Arithmetic coding for data compression. *Communications of the ACM*, 30(6):520–540, 1987. doi: 10.1145/214762.214771.

52. Maciej F. Boni, Philippe Lemey, Xiaowei Jiang, Tommy Tsan-Yuk Lam, Blair W. Perry, Todd A. Castoe, Andrew Rambaut, and David L. Robertson. Evolutionary origins of the SARS-CoV-2 sarbecovirus lineage responsible for the COVID-19 pandemic. *Nature Microbiology*, 5(11):1408–1417, 2020. doi: 10.1038/s41564-020-0771-4.

53. Spyros Lytras, Joseph Hughes, Darren Martin, Phillip Swanepoel, Alice de Klerk, Rentia Lourens, Sergei L. Kosakovsky Pond, Wei Xia, Xiaowei Jiang, and David L. Robertson.

Exploring the natural origins of SARS-CoV-2 in the light of recombination. *Genome Biology and Evolution*, 14(2):evac018, 2022. doi: 10.1093/gbe/evac018.

54. Donovan H. Parks, Maria Chuvochina, David W. Waite, Christian Rinke, Adam Skarshewski, Pierre-Alain Chaumeil, and Philip Hugenholtz. A standardized bacterial taxonomy based on genome phylogeny substantially revises the tree of life. *Nature Biotechnology*, 36(10):996–1004, 2018. doi: 10.1038/nbt.4229.

55. Carl R. Woese, Gary J. Olsen, Michael Ibba, and Dieter Söll. Aminoacyl-tRNA synthetases, the genetic code, and the evolutionary process. *Microbiology and Molecular Biology Reviews*, 64(1):202–236, 2000. doi: 10.1128/MMBR.64.1.202-236.2000.

56. Kelly L. Wyres, Margaret M. C. Lam, and Kathryn E. Holt. Population genomics of *Klebsiella pneumoniae*. *Nature Reviews Microbiology*, 18(6):344–359, 2020. doi: 10.1038/s41579-019-0315-1.

57. Cindy J. Castelle and Jillian F. Banfield. Major new microbial groups expand diversity and alter our understanding of the tree of life. *Cell*, 172(6):1181–1197, 2018. doi: 10.1016/j.cell.2018.02.016.

58. Daniel McDonald, Morgan N. Price, Julia Goodrich, Eric P. Nawrocki, Todd Z. DeSantis, Alexander Probst, Gary L. Andersen, Rob Knight, and Philip Hugenholtz. An improved Greengenes taxonomy with explicit ranks for ecological and evolutionary analyses of bacteria and archaea. *The ISME Journal*, 6(3):610–618, 2012. doi: 10.1038/ismej.2011.139.

59. Jeroen M. J. Tas, Luka Mesin, Giulia Pasqual, Sasha Targ, Johanne T. Jacobsen, Yasuko M. Mano, Casie S. Chen, Jean-Claude Weill, Claude-Agnès Reynaud, Edward P. Browne, Michael Meyer-Hermann, and Gabriel D. Victora. Visualizing antibody affinity maturation in germinal centers. *Science*, 351(6277):1048–1054, 2016. doi: 10.1126/science.aad3439.

60. William S. DeWitt, Luka Mesin, Gabriel D. Victora, Vladimir N. Minin, and Frederick A. Matsen, IV. Using genotype abundance to improve phylogenetic inference. *Molecular Biology and Evolution*, 35(5):1253–1265, 2018. doi: 10.1093/molbev/msy020.

61. Etienne Sollier, Jack Kuipers, Koichi Takahashi, Niko Beerenwinkel, and Katharina Jahn. COMPASS: joint copy number and mutation phylogeny reconstruction from amplicon single-cell sequencing data. *Nature Communications*, 14(1):4921, 2023. doi: 10.1038/s41467-023-40378-8.

62. Sohrab Salehi, Fatemeh Dorri, Kevin Chern, Farhia Kabeer, Nicole Rusk, Tyler Funnell, Marc J. Williams, Daniel Lai, Mirela Andronescu, Kieran R. Campbell, Andrew McPherson, Samuel Aparicio, Andrew Roth, Sohrab P. Shah, and Alexandre Bouchard-Côté. Cancer phylogenetic tree inference at scale from 1000s of single cell genomes. *Peer Community Journal*, 3:e63, 2023. doi: 10.24072/pcjournal.292.

63. Hirotugu Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974. doi: 10.1109/TAC.1974.1100705.

64. Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978. doi: 10.1214/aos/1176344136.

65. Subha Kalyaanamoorthy, Bui Quang Minh, Thomas K. F. Wong, Arndt von Haeseler, and Lars S. Jermiin. ModelFinder: fast model selection for accurate phylogenetic estimates. *Nature Methods*, 14(6):587–589, 2017. doi: 10.1038/nmeth.4285.

66. George E. P. Box. Science and statistics. *Journal of the American Statistical Association*, 71(356):791–799, 1976. doi: 10.1080/01621459.1976.10480949.

67. Edward Susko and Andrew J. Roger. On the use of information criteria for model selection in phylogenetics. *Molecular Biology and Evolution*, 37(2):549–562, 2020. doi: 10.1093/molbev/msz228.

68. Maria Anisimova, Manuel Gil, Jean-François Dufayard, Christophe Dessimoz, and Olivier Gascuel. Survey of branch support methods demonstrates accuracy, power, and robustness of fast likelihood-based approximation schemes. *Systematic Biology*, 60(5):685–699, 2011. doi: 10.1093/sysbio/syr041.

69. Diep Thi Hoang, Olga Chernomor, Arndt von Haeseler, Bui Quang Minh, and Le Sy Vinh. UFBoot2: improving the ultrafast bootstrap approximation. *Molecular Biology and Evolution*, 35(2):518–522, 2018. doi: 10.1093/molbev/msx281.

70. Li Jin and Masatoshi Nei. Limitations of the evolutionary parsimony method of phylogenetic analysis. *Molecular Biology and Evolution*, 7(1):82–102, 1990. doi: 10.1093/oxfordjournals.molbev.a040588.

71. Stéphane Guindon, Jean-François Dufayard, Vincent Lefort, Maria Anisimova, Wim Hordijk, and Olivier Gascuel. New algorithms and methods to estimate maximum-likelihood phylogenies: assessing the performance of PhyML 3.0. *Systematic Biology*, 59(3):307–321, 2010. doi: 10.1093/sysbio/syq010.

72. David F. Robinson and Leslie R. Foulds. Comparison of phylogenetic trees. *Mathematical Biosciences*, 53(1-2):131–147, 1981. doi: 10.1016/0025-5564(81)90043-2.

73. Zhenxiang Xi, Liang Liu, Joshua S. Rest, repo, and Charles C. Davis. Coalescent versus concatenation methods and the placement of *Amborella* as sister to water lilies. *Systematic Biology*, 63(6):919–932, 2014. doi: 10.1093/sysbio/syu055.

74. Pascal-Antoine Christin, Nicolas Salamin, Vincent Savolainen, Melvin R. Duvall, and Guillaume Besnard. C_4 photosynthesis evolved in grasses via parallel adaptive genetic changes. *Current Biology*, 17(14):1241–1247, 2007. doi: 10.1016/j.cub.2007.06.036.

75. Luke T. Dunning, Jill K. Olofsson, Christian Parisod, Rimjhim Roy Choudhury, Jose J. Moreno-Villena, Yang Yang, Jacqueline Dionora, W. Paul Quick, Mi Park, Jeffrey L. Bennetzen, et al. Lateral transfers of large DNA fragments spread functional genes among grasses. *Proceedings of the National Academy of Sciences*, 116(10):4416–4425, 2019. doi: 10.1073/pnas.1810031116.

76. Nicholas J. Croucher, Simon R. Harris, Claire Fraser, Michael A. Quail, Jonathan Burton, Mark van der Linden, Lesley McGee, Anne von Gottberg, Jae Hoon Song, Kwan Soo Ko, et al. Rapid pneumococcal evolution in response to clinical interventions. *Science*, 331(6016):430–434, 2011. doi: 10.1126/science.1198545.

77. David L. Robertson, J. P. Anderson, J. A. Bradac, Jean K. Carr, B. Foley, R. K. Funkhouser, Feng Gao, Beatrice H. Hahn, M. L. Kalish, Carla Kuiken, et al. HIV-1 nomenclature proposal. *Science*, 288(5463):55–56, 2000. doi: 10.1126/science.288.5463.55d.

78. Carolina Pardo-Diaz, Camilo Salazar, Simon W. Baxter, Claire Merot, W. Figueiredo-Ready, Mathieu Joron, W. Owen McMillan, and Chris D. Jiggins. Adaptive introgression across species boundaries in *Heliconius* butterflies. *PLoS Genetics*, 8(6):e1002752, 2012. doi: 10.1371/journal.pgen.1002752.

79. Heliconius Genome Consortium. Butterfly genome reveals promiscuous exchange of mimicry adaptations among species. *Nature*, 487(7405):94–98, 2012. doi: 10.1038/nature11041.

80. Jennifer Ripplinger and Jack Sullivan. Does choice in model selection affect maximum likelihood analysis? *Systematic Biology*, 57(1):76–85, 2008. doi: 10.1080/10635150801898920.

81. Stephanie J. Spielman. Relative model fit does not predict topological accuracy in single-gene protein phylogenetics. *Molecular Biology and Evolution*, 37(7):2110–2123, 2020. doi: 10.1093/molbev/msaa075.

82. James E. Tarver, Mario dos Reis, Siavash Mirarab, Raymond J. Moran, Sean Parker, Joseph E. O'Reilly, Benjamin L. King, Mary J. O'Connell, Robert J. Asher, Tandy Warnow, Kevin J. Peterson, Philip C. J. Donoghue, and Davide Pisani. The interdependence of molecular clock models and tree topologies in the estimation of placental mammal divergence times. *Genome Biology and Evolution*, 8(2):330–344, 2016. doi: 10.1093/gbe/evv261.

83. Chao Zhang, Maryam Rabiee, Erfan Sayyari, and Siavash Mirarab. ASTRAL-III: polynomial time species tree reconstruction from partially resolved gene trees. *BMC Bioinformatics*, 19 (Suppl 6):153, 2018. doi: 10.1186/s12859-018-2129-y.

84. Kalina T. J. Davies, James A. Cotton, John D. Kirwan, Emma C. Teeling, and Stephen J. Rossiter. Parallel signatures of sequence evolution among hearing genes in echolocating mammals: an emerging model of genetic convergence. *Heredity*, 108(5):480–489, 2012. doi: 10.1038/hdy.2011.119.

85. Yong-Yi Shen, Lu Liang, Guo-Song Li, Robert W. Murphy, and Ya-Ping Zhang. Parallel evolution of auditory genes for echolocation in bats and toothed whales. *PLoS Genetics*, 8 (6):e1002788, 2012. doi: 10.1371/journal.pgen.1002788.

86. Christopher G. Dowson, A. Hutchison, J. A. Brannigan, R. C. George, D. Hansman, J. Liñares, A. Tomasz, J. Maynard Smith, and Brian G. Spratt. Horizontal transfer of penicillin-binding protein genes in penicillin-resistant clinical isolates of *Streptococcus pneumoniae*. *Proceedings of the National Academy of Sciences*, 86(22):8842–8846, 1989. doi: 10.1073/pnas.86.22.8842.

87. Na Li and Matthew Stephens. Modeling linkage disequilibrium and identifying recombination hotspots using single-nucleotide polymorphism data. *Genetics*, 165(4):2213–2233, 2003. doi: 10.1093/genetics/165.4.2213.